

Model for Simulating Aerosol Interactions and Chemistry (MOSAIC)

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Received 11 April 2007; revised 20 January 2008; accepted 8 February 2008; published 3 July 2008.

[1] This paper describes and evaluates a new Model for Simulating Aerosol Interactions and Chemistry (MOSAIC), with a special focus on addressing the long-standing issues in solving the dynamic partitioning of semivolatile inorganic gases (HNO_3 , HCl , and NH_3) to size-distributed atmospheric aerosol particles. The coupled ordinary differential equations (ODE) for dynamic gas-particle mass transfer are extremely stiff, and the available numerical techniques are either very expensive or produce oscillatory solutions. These limitations are overcome in MOSAIC with a new dynamic gas-particle partitioning module, which is coupled to an efficient and accurate thermodynamics module. The algorithm includes a new concept of “dynamic pH,” a novel formulation for mass transfer to mixed-phase and solid particles, and an adaptive time stepping scheme, which together hold the key to smooth, accurate, and efficient solutions of gas-particle partitioning over the entire relative humidity range. MOSAIC is found to be in excellent agreement with a benchmark version of the model that uses a rigorous solver for integrating the stiff ODEs. The steady-state MOSAIC results for monodisperse aerosol test cases are also in excellent agreement with those obtained with the benchmark equilibrium model AIM. Moreover, the CPU times required for fully dynamic solutions by MOSAIC per size bin per 5 min intervals (typical 3-D model time steps) are similar to those for bulk equilibrium solutions by the computationally efficient but relatively less accurate model ISORROPIA. These results show that MOSAIC is extremely efficient without compromising accuracy, and is therefore highly attractive for use in air quality and regional/global aerosol models.

Citation: Zaveri, R. A., R. C. Easter, J. D. Fast, and L. K. Peters (2008), Model for Simulating Aerosol Interactions and Chemistry (MOSAIC), *J. Geophys. Res.*, 113, D13204, doi:10.1029/2007JD008782.

1. Introduction

[2] Atmospheric aerosols are ubiquitous suspended particulate matter that can range from a few nanometers up to a few microns in size. In large urban areas and megacities, anthropogenic aerosols are implicated in many human health related problems [Gauderman *et al.*, 2000; Osornio-Vargas *et al.*, 2003; Pope *et al.*, 2004]. On regional and global scales, these as well as other naturally occurring aerosols play an important role in the Earth’s radiative balance and climate via various direct and indirect effects [Intergovernmental Panel on Climate Change, 2001], and are also involved in other environmental issues such as visibility degradation [Malm *et al.*, 1994; Park *et al.*, 2006] and acid deposition [Norris *et al.*, 1999; Kelly *et al.*, 2002]. Aerosols also provide reactive volumes and surfaces in the atmosphere where heterogeneous chemical reactions can proceed efficiently [Laskin *et al.*, 2005; Gibson *et al.*, 2006; Brown *et al.*, 2006]. These interlinked scientific and

policy-related issues have motivated many field, laboratory, and theoretical studies with a common goal of developing a better understanding of the composition of atmospheric aerosols and how they form and evolve as a function of time and space. To this end, several different aerosol process modules and 3-D models have played a central role in first synthesizing the diverse experimental findings with theoretical formulations, and subsequently in interpreting field observations and assessing the impact of aerosols on air quality and climate at different spatial and temporal scales [Peters *et al.*, 1995; Haywood and Boucher, 2000; Zhang *et al.*, 2004; Textor *et al.*, 2006].

[3] However, aerosol modeling is a scientifically challenging endeavor as well as a computationally difficult problem. Atmospheric aerosols not only span over three orders of magnitude in size, but they are also physically and chemically rather complex. Field observations show that they can be composed of a wide variety of compounds, including sulfate, nitrate, ammonium, carbonaceous materials, sea salt, and crustal species from soil dust [Seinfeld and Pandis, 1998]. Furthermore, carbonaceous aerosols include sooty particles (black carbon) formed as result of fossil fuel combustion and biomass burning as well as organic compounds of both anthropogenic and biogenic origins. Particles that are directly emitted into the atmosphere are called

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primary aerosols, which include sea salt, soil dust, black carbon, and organic carbon. However, a significant portion of aerosol mass is formed via the process of gas-particle partitioning of H_2SO_4 , HNO_3 , HCl , and NH_3 , and myriad low and semivolatile organic compounds. The resulting particulate mass, referred to as secondary aerosols, consists of various salts of sulfate, nitrate, chloride, and ammonium and many different organic species. Other important chemical and microphysical processes that affect aerosol size, number, and composition distributions include homogeneous nucleation (new particle formation), coagulation, gas-phase chemistry, heterogeneous chemistry, cloud droplet and ice particle nucleation, dry-deposition, and wet removal.

[4] A reliable aerosol model must therefore be able to resolve the wide particle size range and chemical complexity arising from the many different primary and secondary aerosol species; and it must also include reliable treatments for simulating the various chemical and microphysical processes mentioned above. While great advances have been made on developing model representations of these processes, several microphysical processes such as homogeneous, cloud droplet, and ice particle nucleation and gas-particle partitioning are still somewhat problematic. The difficulties with the different types of nucleation processes and partitioning of organic species are at a fundamental level where the physics and chemistry of the phenomena under ambient atmospheric conditions still remain to be completely understood [e.g., *Nadykto and Yu*, 2007; *Curtius et al.*, 2006; *Cantrell et al.*, 2001; *Stroud et al.*, 2007; *Cziczo et al.*, 2004; *Robinson et al.*, 2007]. On the other hand, the theoretical formulation of the gas-particle partitioning process for inorganic species has been well established. The difficulty primarily lies in its numerical solution.

[5] In this paper, we describe the development and evaluation of a new aerosol model, referred to as Model for Simulating Aerosol Interactions and Chemistry (MOSAIC), with a focus on resolving the outstanding issues associated with the gas-particle partitioning process for the various inorganic gases. For other aerosol processes, we rely on the techniques and modules available in the literature. We begin in section 2 with a brief description of the overall framework of MOSAIC which serves as a host for coupling various chemical and microphysical processes. In section 3, we review the gas-partitioning problem and the different approaches and numerical techniques developed so far. In section 4, we describe in detail the new gas-particle mass transfer module ASTEM (Adaptive Step Time-Split Euler Method) that resolves the outstanding issues with the gas-particle partitioning problem. In section 5, we first verify and validate the fidelity of MOSAIC/ASTEM in a box-model framework for several representative test cases, and then apply MOSAIC within PNNL's 3-D Eulerian model, PEGASUS, to evaluate its performance and CPU time requirements under realistic conditions.

[6] An earlier version of MOSAIC was implemented in the chemistry version of the Weather Research and Forecasting (WRF-chem) model [*Grell et al.*, 2005], and successfully applied to study aerosol evolution and direct radiative forcing in the vicinity of Houston during the TexAQS 2000 field campaign [*Fast et al.*, 2006]. The new version described here is now accessible in WRF-chem

and also available in a standalone box-model format upon request.

2. MOSAIC Framework

[7] The framework of MOSAIC is made highly modular to allow easy coupling between various chemical and microphysical processes. The key features and modules implemented in the current version of MOSAIC are described below. In a 3-D model, these chemical and microphysical processes are integrated in a time split fashion with horizontal and vertical transport processes over transport-chemistry interval h_{CHEM} .

2.1. Aerosol Species

[8] As mentioned above, atmospheric aerosols of different types and origins can be composed of a large number of different chemical species. However, because of computer memory and simulation turn-around time constraints, it is not feasible to represent each and every species explicitly in an aerosol model. One must therefore judiciously choose the species based on the desired accuracy for a given application. Keeping this in mind, MOSAIC treats all the major aerosol species important at urban, regional, and global scales. These include sulfate ($\text{SULF} = \text{SO}_4^{2-} + \text{HSO}_4^-$), methanesulfonate (CH_3SO_3), nitrate (NO_3), chloride (Cl), carbonate (CO_3), ammonium (NH_4), sodium (Na), calcium (Ca), black carbon (BC), primary organic mass (OC), and liquid water (W). Other unspecified inorganic species such as silica (SiO_2), other inert minerals, and trace metals are lumped together as “other inorganic mass” (OIN). Furthermore, reactive inorganic species such as potassium (K) and magnesium (Mg) are not explicitly modeled in MOSAIC as they are usually present in much smaller amounts compared to other cations in aerosols. Instead, K and Mg are represented by equivalent amounts of Na since their sulfate, nitrate, and chloride salts are similar in terms of their solubility in water. The gas-phase species that are allowed to partition to the particle-phase include H_2SO_4 , HNO_3 , HCl , NH_3 , and MSA (methanesulfonic acid). Work is currently underway to include secondary organic aerosols (SOA).

2.2. Gas-Phase Chemistry

[9] The urban to global scale trace gas chemistry in MOSAIC is presently modeled with the “lumped-structure” photochemical mechanism CBM-Z [*Zaveri and Peters*, 1999], which affords a reasonable tradeoff between accuracy and computational efficiency. CBM-Z is based on the widely used Carbon Bond Mechanism (CBM-IV) developed by *Gery et al.* [1989] for use in urban air-shed models. It extends the original mechanism to function properly at regional to global spatial scales and longer time periods than the typical urban air-shed simulations. A condensed dimethylsulfide (DMS) photooxidation mechanism [*Zaveri*, 1997] has also been added to simulate the temperature dependent formation of SO_2 , H_2SO_4 , and MSA in the marine environment. The coupled nonlinear ordinary differential equations describing the gas-phase reactions are solved using the RODAS3 solver [*Sandu et al.*, 1997].

[10] The augmented CBM-Z scheme consists of 67 prognostic species and 164 reactions. However, depending on

the application and simulation domain, all species and reactions may not be active at all the grid cells all the time. We have therefore implemented the entire mechanism in a regime-dependent form in which the species and reactions are partitioned into four submechanisms: (1) background (32 species and 74 reactions), (2) urban (19 species and 44 reactions), (3) biogenic (5 species and 16 reactions), and (5) DMS marine (11 species and 30 reactions). In this approach, the background mechanism is always active while other submechanisms are invoked only when one or more species in that submechanism is present above a threshold concentration of 1 pptv. This approach reduces the CPU time for gas-phase chemistry by up to 30% compared to using the full mechanism over the entire 3-D model domain. CBM-Z has been successfully evaluated using smog chamber data [Zaveri and Peters, 1999] as well as comprehensive field measurements collected over urban [Zaveri *et al.*, 2003; Jiang and Fast, 2004; Fast *et al.*, 2006] and regional [Fast *et al.*, 2002; Fast and Heilman, 2005] spatial scales.

2.3. Particle Size-Distribution

[11] MOSAIC can be implemented using either the modal or sectional approach to represent the aerosol size distribution. In the present study, MOSAIC is implemented in the sectional framework where the aerosol size distribution is divided into discrete size bins. The size bins are defined by their lower and upper dry particle diameters, so water uptake or loss does not transfer particles between bins. Furthermore, each bin is assumed to be internally mixed so that all particles within a bin have the same chemical composition, while particles in different bins are externally mixed. The number of bins is flexible and can be specified by the user.

[12] Both mass and number are simulated for each bin. Particle growth or shrinkage resulting from the dynamic gas-particle partitioning of trace gases (H_2SO_4 , $\text{CH}_3\text{SO}_3\text{H}$, HNO_3 , HCl , NH_3 , and eventually secondary organic species) is first calculated in a Lagrangian manner. Transfer of particles between bins is then calculated using the (default) two-moment approach of Simmel and Wurzel [2006] or the moving section approach of Jacobson [1997a].

2.4. Nucleation and Coagulation

[13] Aerosol coagulation is calculated using the algorithm of Jacobson *et al.* [1994] with a Brownian coagulation kernel. Aerosol nucleation is a complex process that likely involves a number of mechanisms and trace species in addition to H_2SO_4 and H_2O , and it continues to be an area of active research. In the boundary layer, binary H_2SO_4 - H_2O and ternary H_2SO_4 - NH_3 - H_2O nucleation parameterizations tend to under-predict and over-predict nucleation rates, respectively [Lucas and Akimoto, 2006], and the revised ternary treatment of Anttila *et al.* [2005] under-predicts nucleation rates. Empirical parameterizations involving condensable organics [e.g., Fan *et al.*, 2006] have performed well in specific modeling applications. Additional modeling approximations arise when the smallest particles represented in the model are larger than the freshly-nucleated particles. Currently, only the Wexler *et al.* [1994] parameterization of H_2SO_4 - H_2O homogeneous nucleation is used in MOSAIC. We expect to add newer, more robust nucleation parameterizations when they become available.

2.5. Thermodynamics and Phase Equilibrium

[14] An aerosol chemistry model requires an accurate thermodynamic module to reliably predict particle deliquescence, water content and solid-liquid phase equilibrium in multicomponent aerosols at a given relative humidity (RH) and temperature (T). It is also needed to compute the mass transfer driving forces for dynamic gas-particle partitioning of various semivolatile species. At the core of the thermodynamic module is another submodule that estimates the activity coefficients of various electrolytes in multicomponent aqueous aerosols. Activity coefficients are repeatedly required in solid-liquid phase equilibrium and gas-particle partitioning calculations. As a result, the thermodynamic module is called several times for each size bin, and usually accounts for a significant fraction of the overall CPU time consumption by the aerosol model. Thus, in addition to being accurate, the thermodynamic module must also be computationally efficient for 3-D modeling applications.

[15] The thermodynamic module in MOSAIC is specially designed for use in dynamic gas-particle partitioning aerosol models, and satisfies both the accuracy and computational efficiency criteria. It consists of the recently developed Multicomponent Taylor Expansion Method (MTEM) to calculate the activity coefficients in aqueous atmospheric aerosols [Zaveri *et al.*, 2005a] and a computationally efficient Multicomponent Equilibrium Solver for Aerosols (MESA) to compute the intraparticle solid-liquid phase equilibrium [Zaveri *et al.*, 2005b]. The salient features of these two submodules are summarized below.

[16] MTEM estimates the mean activity coefficient of an electrolyte in a multicomponent solution on the basis of its values in binary solution of all the electrolytes present in the mixture at the solution water activity, assuming that the water activity (a_w) is equal to the ambient RH. Aerosol water content is calculated using the Zdanovskii-Stokes-Robinson (ZSR) method [Zdanovskii, 1948; Stokes and Robinson, 1966]. For self-consistency most of the MTEM and ZSR parameters are derived using the comprehensive Pitzer-Simonson-Clegg model (PSC) [Pitzer and Simonson, 1986; Clegg *et al.*, 1992] at 298.15 K. MTEM has been evaluated for several multicomponent systems representing various continental and marine aerosols and was found to be appreciably more accurate than the approaches employed by Bromley [1973] and Kusik and Meissner [1978] (KM). MTEM also yields a noniterative solution of the bisulfate ion dissociation in sulfate-rich systems, which is a major computational advantage over other ionic strength based methods that require an expensive iterative solution. MTEM was estimated to be 1–12 times faster than KM and Bromley, and 50 to 500 times faster than PSC when used in a sectional model with eight or more size bins.

[17] When exposed to increasing RH, a dry atmospheric particle composed of two or more soluble salts spontaneously starts absorbing water at the “mutual deliquescence RH” (MDRH) point. As RH increases further, the particle goes through a mixed solid-liquid phase until all the soluble salts are completely dissolved at the saturation RH (SRH). However, once completely deliquesced, the same particle remains in a supersaturated metastable liquid state as the RH is decreased until it spontaneously crystallizes (via homogeneous or heterogeneous nucleation processes) at an RH much below the MDRH [Tang *et al.*, 1995]. This

phenomenon is referred to as hydration hysteresis in atmospheric particles. At any given temperature, the MDRH for a mixed-salt particle depends only on which salts are present while the SRH depends on their relative amounts.

[18] The MESA module in MOSAIC is designed to efficiently and accurately solve the complex multicomponent solid-liquid equilibria within each aerosol size bin when the particles are on the lower curve (i.e., completely solid or mixed-phase) of hydration hysteresis loop [Zaveri *et al.*, 2005b]. It uses a modified pseudotransient continuation technique in which the equilibrium reactions are formulated as pseudotransient precipitation and dissolution reactions and the resulting set of stiff nonlinear ordinary differential equations (ODEs) are integrated until the system reaches a steady state to obtain the equilibrium solution. The stiffness of the ODEs is virtually eliminated by integrating each differential equation using an independent, adaptive time step. Numerically, this approach is equivalent to iterating all the equilibrium equations simultaneously with a single iteration loop. Additionally, an efficient algorithm was developed for estimating temperature dependent MDRH for all the feasible salt mixtures, which eliminated the need for rigorous and expensive numerical solutions when the relative humidity is less than the MDRH for a given salt mixture. Performance of MESA-MTEM was evaluated against the benchmark aerosol thermodynamics model AIM (Web-based Model III) [Wexler and Clegg, 2002] for accuracy and the generalized equilibrium solver EQUISOLV II [Jacobson, 1999] for speed. The MESA-MTEM configuration was not only found to be more accurate than EQUISOLV II, but also 1.4 to 5.8 times faster than EQUISOLV II on average, with a maximum speed ratio of 17. Because MESA directly diagnoses MDRH from a given aerosol composition, it is even faster when the particle-phase is predicted to be completely solid.

[19] As mentioned above, if the particles are on the upper hysteresis curve (i.e., completely liquid) of the hydration hysteresis loop, then they remain in a metastable liquid state as long as the ambient RH is above the crystallization RH (CRH). However, prediction of CRH for any given aerosol composition and temperature is difficult and highly uncertain. Martin *et al.* [2003] found that CRH ranges between 1 and 35% for particles containing ammonium salts of sulfate and nitrate. In the presence of soluble organics and insoluble inclusions like soot and mineral dust, the CRH may either remain unchanged or increase up to 65% [Brooks *et al.*, 2003; Choi and Chan, 2002; Han and Martin, 1999; Martin *et al.*, 2001; Han *et al.*, 2002]. Because of lack of reliable parameterization of CRH as a function of composition and temperature, we have simply assumed a CRH of 35% for all particles containing ammonium and sodium salts of sulfate, nitrate, and chloride, while $\text{Ca}(\text{NO}_3)_2$ and CaCl_2 salts are assumed to crystallize at 10% RH in ambient aerosols, although pure $\text{Ca}(\text{NO}_3)_2$ particles have been observed to remain supersaturated down to less than 1% RH in laboratory experiments [Tang and Fung, 1997]. This treatment will be updated in the future as additional experimental data and more comprehensive parameterizations become available.

[20] Because MSA can be an important aerosol component in marine environments, the published version of the thermodynamic module is extended here to include methanesul-

fonic acid (MSA) and its nonvolatile salts $\text{CH}_3\text{SO}_3\text{NH}_4$, $\text{CH}_3\text{SO}_3\text{Na}$, $(\text{CH}_3\text{SO}_3)_2\text{Ca}$. However, because of the lack of reliable parameterizations of activity coefficients over the entire RH range and the necessary equilibrium constants, these species are assumed to remain in a deliquesced state above 10% RH in the current version of the model. Furthermore, the binary molalities (as a function of water activity) of MSA, $\text{CH}_3\text{SO}_3\text{NH}_4$, $\text{CH}_3\text{SO}_3\text{Na}$, and $(\text{CH}_3\text{SO}_3)_2\text{Ca}$ are assumed to be same as those of HNO_3 , NH_4HSO_4 , NaHSO_4 , and $\text{Ca}(\text{NO}_3)_2$, respectively. This treatment is included for completeness, but will be updated in the future as the relevant experimental data become available.

[21] In the present work, MESA also takes into account the Kelvin Effect for particle surface curvature while computing equilibrium water content in liquid particles by iterating the following equation:

$$\ln \text{RH} = \ln a_{w,m} + \left(\frac{2v_w\sigma_{l,m}}{\mathcal{R}T\bar{R}_{p,m}} \right) \quad (1)$$

where RH (-) is the ambient fractional relative humidity; $a_{w,m}$ is the water activity of particles in size bin m ; $\sigma_{l,m}$ (dyne cm^{-1}) is the liquid-phase surface tension; $\mathcal{R}$ is the universal gas constant; T (K) is the temperature; $\bar{R}_{p,m}$ (cm) is mean wet radius of particles in bin m ; and the partial molar volume of water, $v_w = 18 \text{ cm}^3 \text{ mol}^{-1}$. Presently, we have implemented a simple correlation for estimating surface tension in aqueous salt solutions as a function of temperature and water activity [Zaytsev and Aseyev, 1992]:

$$\sigma_{l,m} = \sigma_w(T) + 49(1 - a_{w,m}) \quad (2)$$

where $\sigma_w(T)$ is the pure water surface tension as a function of temperature. The Kelvin Effect calculation typically requires four iterations or less.

2.6. Hydration Hysteresis

[22] It is relatively straightforward to keep a track of the hydration history of aerosols in a Lagrangian box-model framework. In a 3-D model, however, aerosols in different grid cells with different hydration histories can mix together due to turbulent transport. Since the resulting aerosol size distribution is represented by the same set of bins, all the particles in a given bin must be placed either on the lower curve or on the upper curve of the hydration hysteresis loop. To determine this we transport a variable called “hysteresis water content” (W_{hyst}), which can either be equal to zero if the particles are on the lower curve, or equal to the water content at 60% RH ($W_{60\%}$), assuming complete deliquescence if the particles are on the upper curve. Here 60% RH serves as a reference point that is independent of the ambient RH, which can be different in adjacent grid cells. At the beginning of the transport-chemistry integration interval h_{chem} in a grid cell, if $W_{\text{hyst}} > 0.5W_{60\%}$ for a given aerosol bin, then all the particles in that bin are assumed to be on the upper curve; otherwise, they are assumed to be on the lower curve. At the end of the integration interval, W_{hyst} for a bin on the upper curve is set equal to $W_{60\%}$ while that for a bin on the lower curve is set equal to zero.

2.7. Gas-Particle Partitioning

[23] The new gas-particle partitioning module ASTEM (Adaptive Step Time-Split Euler Method) is coupled here with the thermodynamic module MESA-MTEM to dynamically integrate the mass transfer equations. Before describing the ASTEM algorithm in detail, it would be useful to first discuss the current state of gas-particle partitioning problem by briefly reviewing the different approaches and numerical techniques developed so far, and identify the outstanding issues that still need to be resolved.

3. Gas-Particle Partitioning Problem

[24] Because H_2SO_4 and $\text{CH}_3\text{SO}_3\text{H}$ are effectively non-volatile gases at atmospheric temperatures, their condensation onto preexisting particles is essentially an irreversible kinetic process. On the other hand, gas-particle partitioning of semivolatile species such as HNO_3 , HCl , NH_3 , and secondary organics is largely a reversible phenomenon that is effectively governed by gas-particle mass transfer kinetics, equilibrium thermodynamics, and heterogeneous chemical reactions. These processes are highly sensitive to the ever-changing meteorological controls such as temperature and relative humidity as well as other factors including gas-phase concentrations, particle size distribution, particle composition, phase-state, and pH (for aqueous particles). Thus, gas-particle partitioning of semivolatile species is a highly dynamic and competitive process, and plays an important role in the continuous evolution of aerosols and the associated physical and chemical properties relevant for the climate forcing and air quality issues.

[25] Several investigators have studied the problem of multicomponent gas-particle partitioning since the early 1980s. Some of the early sectional aerosol models predicted the gas-particle partitioning of H_2SO_4 , HNO_3 , HCl , and NH_3 over size- and composition-distributed aerosols assuming thermodynamic equilibrium [Bassett and Seinfeld, 1983; Pilinis *et al.*, 1987; Pilinis and Seinfeld, 1987, 1988]. However, several observational and theoretical studies since then have demonstrated that the equilibrium assumption is not always valid under typical atmospheric conditions [Allen *et al.*, 1989; Wexler and Seinfeld, 1990, 1992]. These studies also indicated that equilibrium alone cannot uniquely determine the distribution of condensed semivolatile gases. Both mass transport and thermodynamics must be considered to accurately predict the size distribution of semivolatile species.

[26] Wexler and Seinfeld [1991] developed one of the first dynamic sectional aerosol models in which the mass transfer rates for semivolatile gases over each aerosol size bin were determined from the intraparticle phase equilibrium calculations and the appropriate gas-particle equilibrium constants. This approach is mathematically represented by a system of nonlinear ordinary differential equations (ODE):

$$\frac{dC_{a,i,m}}{dt} = k_{i,m}(C_{g,i} - C_{a,i,m}^*) \quad (3)$$

$$\frac{dC_{g,i}}{dt} = - \sum_m k_{i,m}(C_{g,i} - C_{a,i,m}^*) \quad (4)$$

where equations (3) and (4) are for the particle- and gas-phase species, respectively; $C_{a,i,m}$ is particle-phase concentration of species i in bin m ; $C_{a,i,m}^*$ is the equilibrium (denoted by $*$) gas-phase concentration of species i with the particle-phase of bin m ; $C_{g,i}$ is the bulk gas-phase concentration of same species i . All concentration variables above have the unit of $\text{mol m}^{-3}(\text{air})$. Lastly, $k_{i,m} (\text{s}^{-1})$ is the first order mass transfer coefficient for species i and bin m , which is estimated as:

$$k_{i,m} = 4\pi\bar{R}_{p,m}D_{g,i}N_m f(Kn_{i,m}, \alpha_i) \quad (5)$$

where $D_{g,i} (\text{cm}^2 \text{s}^{-1})$ is gas diffusivity of species i ; $N_m (\text{cm}^{-3})$ is the number concentration of particles in bin m ; and $f(Kn_{i,m}, \alpha_i)$ is the transition regime correction factor [Fuchs and Sutugin, 1971] to the Maxwellian flux as a function of the Knudsen Number $Kn_{i,m} = \lambda_i/\bar{R}_{p,m}$ (where λ_i is the mean free path):

$$f(Kn_{i,m}, \alpha_i) = \frac{0.75\alpha_i(1 + Kn_{i,m})}{Kn_{i,m}(1 + Kn_{i,m}) + 0.283\alpha_iKn_{i,m} + 0.75\alpha_i} \quad (6)$$

The above correction factor also accounts for interfacial mass transport limitation characterized by the mass accommodation coefficient α_i , which is typically assumed equal to 0.1 for all condensing species for all size bins [Zhang *et al.*, 2004]. Furthermore, the term $(C_{g,i} - C_{a,i,m}^*)$ is called the “driving force” for mass transfer, and the term $k_{i,m}(C_{g,i} - C_{a,i,m}^*)$ is called the “instantaneous flux” ($\text{mol cm}^{-3} \text{s}^{-1}$) of species i to/from bin m .

[27] The system of coupled nonlinear ODEs must be numerically solved to obtain a unique time-dependent solution for all the gas and aerosol species concentrations under all conditions. However, the system of ODEs is very *stiff* and has proved to be extremely difficult to integrate numerically. The stiffness arises because of the large differences in the characteristic times required for the semivolatile species in different aerosol size bins to reach equilibrium, where the characteristic timescale for a species i in bin m is defined as

$$\tau_{i,m} = \frac{C_{a,i,m}}{\left| \frac{dC_{a,i,m}}{dt} \right|} = \frac{C_{a,i,m}}{|k_{i,m}(C_{g,i} - C_{a,i,m}^*)|} \quad (7)$$

Concentrations of species in small size bins representing the ultrafine and Aitken modes are usually much smaller than the accumulation and coarse particle bins. On the other hand, the magnitude of the fluxes of gases over the small size bins is usually higher or similar to that for accumulation and coarse particle bins. As a result, semivolatile species in the smaller size bins have much shorter characteristic times compared to the larger size bins. This is especially problematic during evaporation of solid-phase species with very small concentrations. Additional stiffness and non-linearity is introduced by H^+ ions in completely or partially liquid aerosol particles. Fluxes of HNO_3 , HCl , and NH_3 to the liquid phase not only depend on the H^+ ion concentration (via C^*), but their sum, along with other acid-base species, also governs its instantaneous value. This leads to even shorter characteristic times for H^+ ions, especially in “acid-neutral” aerosols where the pseudosteady-state con-

concentrations of H^+ ions are two or more orders of magnitude smaller than the sum of all the cations [Sun and Wexler, 1998]. Since semivolatile species in all the aerosol bins are coupled together via the gas phase, a given numerical integrator must take time steps on the order of the shortest timescale to ensure adequate accuracy for all the species in all the bins across the entire aerosol size distribution. These are the reasons why dynamic aerosol models are computationally very expensive.

[28] Several investigators have approached this problem with different techniques to reduce the stiffness of the system of ODEs so that it could be solved efficiently. Jacobson [1997b] derived a semi-implicit exponential numerical scheme that appears to reliably solve the mass transfer equations using 5 to 15 s time steps. However, longer time steps, especially at low RH, induced oscillations in concentrations due to significant delays in feedback between the intraparticle phase equilibrium and mass transfer calculations. This was partly because the scheme relied on the assumption of small amounts of “artificial” liquid water for solving the growth equations even when the aerosol would be completely dry at the given RH according to thermodynamic equilibrium. Jacobson [2002] tried to address this issue by solving transfer of HNO_3 , HCl , and NH_3 as a condensation process (instead of dissolution) for size bins where solid NH_4NO_3 and NH_4Cl already existed.

[29] As mentioned earlier, another source of stiffness and instability in the numerical solution is the independent transfer of the acids and NH_3 between the gas and aerosol phases. This causes the H^+ ion concentration to oscillate when the time steps are greater than τ_{H^+} . To overcome this problem, Sun and Wexler [1998] introduced the concept of coupled condensation in which the NH_3 flux is forced to be exactly equal to the sum of all the acid fluxes (HNO_3 , HCl , and H_2SO_4) when the aerosol is near acid neutrality. While this approach eliminates oscillations in H^+ ion, it does not always allow for gradual changes in the pH as the gas- and particle-phase species concentrations change with time, and may therefore lead to inaccurate steady-state H^+ ion concentrations. In an attempt to rectify this problem, Pilinis *et al.* [2000] relaxed the coupled condensation constraint, and instead, artificially restricted the net H^+ flux to a “critical acidity” flux value that is allowed to vary proportionally with the aerosol H^+ ion concentration. They also derived an extensive set of criteria and equations for solving irreversible and reversible gas-particle reactions over solid aerosols of different compositions, a problem that other models in the past had neglected. The coupled system of ODEs was solved using the double precision version of LSODE (Livermore Solver for Ordinary Differential Equations), a robust stiff-ODE solver [Hindmarsh, 1983]. However, they did not compare their time-varying solutions against a pure equilibrium solution for a monodisperse aerosol to verify whether the artificial H^+ flux constraint actually produces accurate steady-state solutions. Solving gas-particle mass transfer with LSODE is computationally too expensive for use in 3-D models. Therefore, to reduce the computational cost, Capaldo *et al.* [2000] developed a “hybrid” method that utilizes the equilibrium assumption for the fine aerosol mode (i.e., particles with diameters less than $1\ \mu m$) and the dynamic approach for the coarse aerosol mode. Koo *et al.* [2003] evaluated the bulk equilibrium, dynamic, and hybrid

methods, and found that the hybrid method was only 2–3 times faster than the fully dynamic method, but almost 50 times slower than the bulk equilibrium approach. In any case, the hybrid approach itself is problematic because it lumps all the fine mode bins together when calculating gas-particle equilibrium, thereby ignoring and disallowing any compositional and phase-state differences among all the different fine-mode size bins. Also, while the hybrid approach may appear to work well in a polluted urban scenario in the first few hours, it does not capture the continued evolution of the aerosol size and composition distribution due to the slow transfer of semivolatile species among the deliquesced submicron particles via the gas phase over longer timescales [Wexler and Seinfeld, 1992].

[30] More recently, Jacobson [2005] revised his semi-implicit scheme to address the previous issues that produced oscillatory solutions at longer time steps. In the updated scheme, transfer of HNO_3 and HCl is treated as condensational growth rather than dissolution process at low RH (corresponding to total aerosol liquid water content (LWC) below $0.01\ \mu g\ m^{-3}$), provided the appropriate gas-solid equilibrium constraints involving NH_3 are satisfied. However, dynamic mass transfer is solved for the acids only, while NH_3 is added to each size bin at the end of each time step by equilibrating it over all size bins for all LWCs, which requires up to 12 or more iterations. This is followed by another set of nested iterations for more completely updating the solid-liquid phase equilibrium and pH. These several layers of iterations can prove to be rather expensive. Furthermore, while the dynamic solutions at long time steps (150–300 s) appear to match the equilibrium results after several hours, they still do contain oscillations, especially at moderate RH where solids and liquids coexisted. Also, for three out of the six test cases shown in the paper, the long time step results do not appear to agree well with the dynamic solutions using 5 s steps during the first 30 min or more. It is important to obtain a reasonably accurate solution in the initial period, because most 3-D models employ a transport-chemistry time splitting interval h_{CHEM} of 5 to 30 min. The new asynchronous time stepping (ATS) integrator for aerosol dynamics developed by Zhang and Wexler [2006] appears to be promising. In this technique, the solver groups particulate species by particle sections and their characteristic timescales and reduces the numbers of integrations for the long time-scale (“slow”) variables compared to the short time-scale (“fast”) variables to speed up computations.

[31] While significant progress has been made so far, there still remain at least two unresolved issues: (1) lack of an accurate and efficient method to estimate the H^+ ion concentration for calculating fluxes to/from liquid particles; and (2) lack of an accurate and efficient algorithm for solving the dynamic gas-particle partitioning at low and moderate RHs when aerosol particles may be completely solid or mixed-phase. Here we describe the new ASTEM algorithm that addresses these issues and produces accurate, smooth, and computationally efficient solutions at low, moderate, and high RHs.

4. ASTEM Algorithm

[32] As stated earlier, ASTEM stands for Adaptive Step Time-split Euler Method. It is specially designed to take

Table 1. List of Irreversible Heterogeneous Reactions Included in MOSAIC

No.	Irreversible Heterogeneous Reactions
<i>Reactions With H₂SO₄(g)</i>	
(R1)	$\text{CaCO}_3(s) + \text{H}_2\text{SO}_4(g) \rightarrow \text{CaSO}_4(s) + \text{H}_2\text{O}(g) \uparrow + \text{CO}_2(g) \uparrow$
(R2)	$\text{CaCl}_2(s, l) + \text{H}_2\text{SO}_4(g) \rightarrow \text{CaSO}_4(s) + 2\text{HCl}(g) \uparrow$
(R3)	$\text{Ca}(\text{NO}_3)_2(s, l) + \text{H}_2\text{SO}_4(g) \rightarrow \text{CaSO}_4(s) + 2\text{HNO}_3(g) \uparrow$
(R4)	$2\text{NaCl}(s, l) + \text{H}_2\text{SO}_4(g) \rightarrow \text{Na}_2\text{SO}_4(s, l) + 2\text{HCl}(g) \uparrow$
(R5)	$2\text{NaNO}_3(s, l) + \text{H}_2\text{SO}_4(g) \rightarrow \text{Na}_2\text{SO}_4(s, l) + 2\text{HNO}_3(g) \uparrow$
(R6)	$(\text{CH}_3\text{SO}_3)_2\text{Ca}(s, l) + \text{H}_2\text{SO}_4(g) \rightarrow \text{CaSO}_4(s) + 2\text{CH}_3\text{SO}_3\text{H}(l)$
<i>Reactions With CH₃SO₃H(g)</i>	
(R7)	$\text{CaCO}_3(s) + 2\text{CH}_3\text{SO}_3\text{H}(g) \rightarrow (\text{CH}_3\text{SO}_3)_2\text{Ca}(s, l) + \text{H}_2\text{O}(g) \uparrow + \text{CO}_2(g) \uparrow$
(R8)	$\text{CaCl}_2(s, l) + 2\text{CH}_3\text{SO}_3\text{H}(g) \rightarrow (\text{CH}_3\text{SO}_3)_2\text{Ca}(s, l) + 2\text{HCl}(g) \uparrow$
(R9)	$\text{Ca}(\text{NO}_3)_2(s, l) + 2\text{CH}_3\text{SO}_3\text{H}(g) \rightarrow (\text{CH}_3\text{SO}_3)_2\text{Ca}(s, l) + 2\text{HNO}_3(g) \uparrow$
(R10)	$\text{NaCl}(s, l) + \text{CH}_3\text{SO}_3\text{H}(g) \rightarrow \text{CH}_3\text{SO}_3\text{Na}(s, l) + \text{HCl}(g) \uparrow$
(R11)	$\text{NaNO}_3(s, l) + \text{CH}_3\text{SO}_3\text{H}(g) \rightarrow \text{CH}_3\text{SO}_3\text{Na}(s, l) + \text{HNO}_3(g) \uparrow$
<i>Reactions With HNO₃(g)</i>	
(R12)	$\text{CaCO}_3(s) + 2\text{HNO}_3(g) \rightarrow \text{Ca}(\text{NO}_3)_2(s) + \text{H}_2\text{O}(g) \uparrow + \text{CO}_2(g) \uparrow$
(R13)	$\text{CaCl}_2(s) + 2\text{HNO}_3(g) \rightarrow \text{Ca}(\text{NO}_3)_2(s) + 2\text{HCl}(g) \uparrow$
(R14)	$\text{NaCl}(s) + \text{HNO}_3(g) \rightarrow \text{NaNO}_3(s) + \text{HCl}(g) \uparrow$
<i>Reactions With HCl(g)</i>	
(R15)	$\text{CaCO}_3(s) + 2\text{HCl}(g) \rightarrow \text{CaCl}_2(s) + \text{H}_2\text{O}(g) \uparrow + \text{CO}_2(g) \uparrow$
<i>Reactions With NH₃(g)</i>	
(R16)	$\text{NH}_4\text{HSO}_4(s) + \text{NH}_3(g) \rightarrow (\text{NH}_4)_2\text{SO}_4(s)$
(R17)	$(\text{NH}_4)_3\text{H}(\text{SO}_4)_2(s) + \text{NH}_3(g) \rightarrow 2(\text{NH}_4)_2\text{SO}_4(s)$
(R18)	$2\text{NaHSO}_4(s) + \text{NH}_3(g) \rightarrow \text{Na}_2\text{SO}_4(s) + \text{NH}_4\text{HSO}_4(s)$

advantage of several prominent characteristics of the atmospheric gas-particle partitioning problem to systematically reduce the stiffness while still maintaining the overall fidelity of the numerical solution. The first step in reducing the stiffness involves separating the nonvolatile gases from semivolatile ones in the numerical solver. Because gases such as H₂SO₄ and MSA are nonvolatile, their solution is relatively straightforward and does not depend on the phase state of the particles in different bins. On the other hand, mass transfer of semivolatile HNO₃, HCl, and NH₃ is rather complex and depends greatly on the phase-state of the aerosols. Thus, ASTEM consists of two parts which are time-split. In the first part, ASTEM analytically integrates the condensation of nonvolatile gases H₂SO₄ and MSA for all the bins over a user-specified time splitting interval, h_{ASTEM} , typically set at 5 min. In the second part, it numerically integrates condensation and/or evaporation of HNO₃, HCl, and NH₃ for all the bins over h_{ASTEM} using adaptive time steps, h . The two parts are described below.

4.1. Condensation of H₂SO₄ and MSA

[33] Since H₂SO₄ and MSA are nonvolatile, it is reasonable to assume their equilibrium surface concentrations to be zero. Furthermore, by assuming that the mass transfer coefficients of H₂SO₄ and MSA for all the bins remain constant over h_{ASTEM} , the condensation of these nonvolatile gases reduces to a simple first order process:

$$\frac{dC_{g,i}}{dt} = -\left(\sum_m k_{i,m}\right)C_{g,i}. \quad (8)$$

A trivial analytical integration the above equation from the “old time,” t , to the “new time,” $t + h_{\text{ASTEM}}$, gives the new gas-phase concentration as:

$$C_{g,i}^{t+h_{\text{ASTEM}}} = C_{g,i}^t \exp\left(-h_{\text{ASTEM}} \times \sum_m k_{i,m}^t\right). \quad (9)$$

The new particle-phase concentrations in each bin are then calculated from:

$$C_{a,i,m}^{t+h_{\text{ASTEM}}} = C_{a,i,m}^t + \left(C_{g,i}^t - C_{g,i}^{t+h_{\text{ASTEM}}}\right) \frac{k_{i,m}^t}{\sum_m k_{i,m}^t}. \quad (10)$$

[34] If H₂SO₄ and MSA condense on particles containing any one or mixture of salts such as CaCO₃, Ca(NO₃)₂, CaCl₂, NaNO₃, and NaCl, then the pertinent irreversible heterogeneous ((R1) through (R11)) listed in Table 1 are invoked and the appropriate amounts of CO₂, HCl, and HNO₃ gases are evaporated. However, if any of these salts are not present in the preexisting aerosol bins, then some NH₃ is allowed to condense so that the aerosols do not artificially become “sulfate rich” before HNO₃, HCl, and additional NH₃ are condensed or evaporated in the next part. The amount of NH₃ allowed to condense in this part is estimated as:

$$C_{a,\text{NH}_4,m}^{t+h_{\text{ASTEM}}} = C_{a,\text{NH}_4,m}^t + \min\left(\left(2\Delta C_{a,\text{SULF},m}^{t+h_{\text{ASTEM}}} + \Delta C_{a,\text{CH}_3\text{SO}_3,m}^{t+h_{\text{ASTEM}}}\right), \Delta C_{a,\text{NH}_4,m}^{\max}\right) \quad (11)$$

where

$$\Delta C_{a,\text{SULF},m}^{t+h,\text{ASTEM}} = C_{a,\text{SULF},m}^{t+h,\text{ASTEM}} - C_{a,\text{SULF},m}^t, \quad (12)$$

$$\Delta C_{a,\text{CH}_3\text{SO}_3,m}^{t+h,\text{ASTEM}} = C_{a,\text{CH}_3\text{SO}_3,m}^{t+h,\text{ASTEM}} - C_{a,\text{CH}_3\text{SO}_3,m}^t, \quad \text{and} \quad (13)$$

$$\Delta C_{a,\text{NH}_4,m}^{\text{max}} = C_{g,\text{NH}_3}^t \left[1 - \exp \left(-h_{\text{ASTEM}} \times \sum_m k_{\text{NH}_3,m}^t \right) \right] \quad (14)$$

Once the new particle-phase sulfate (SULF), CH_3SO_3 , and NH_4 concentrations are computed, it is necessary to update the internal solid-liquid phase equilibrium in each size bin, which is done by the thermodynamic module MESA. The “dry-state” salt formation hierarchies in MESA [Zaveri *et al.*, 2005b, Table 4, Table 5] are first used to compute the new total particle-phase salt concentrations, and then the moles of HNO_3 and HCl that are displaced due to H_2SO_4 and MSA condensation are instantly evaporated. MESA then partitions all the total particle-phase species concentrations ($C_{a,i,m}$) into the solid-phase ($C_{s,i,m}$) and liquid-phase ($C_{l,i,m}$) concentrations. These are then input to the next part of the ASTEM algorithm to compute the dynamic mass transfer of HNO_3 , HCl , and NH_3 .

4.2. Condensation/Evaporation of HNO_3 , HCl and NH_3

[35] As mentioned earlier, the gas-particle mass transfer rates of HNO_3 , HCl , and NH_3 greatly depend on whether the particles are completely solid, completely liquid, or mixed phase. It is then necessary to explicitly track the aerosol species concentrations in both the solid and liquid phases during each time step taken to integrate HNO_3 , HCl , and NH_3 over the h_{ASTEM} time interval. Therefore, instead of solving equation (3) for the total particle-phase concentrations, ASTEM solves two equations – one for each particle-phase:

$$\frac{dC_{l,i,m}}{dt} = k_{i,m} H_{l,i,m} (C_{g,i} - C_{l,i,m}^*), \quad (15)$$

$$\frac{dC_{s,i,m}}{dt} = k_{i,m} H_{s,i,m} (C_{g,i} - C_{s,i,m}^*). \quad (16)$$

Here $i = \text{HNO}_3$, HCl , and NH_3 , and $C_{l,i,m}^*$ and $C_{s,i,m}^*$ are the equilibrium concentrations for the liquid and solid phases, respectively. Also note that Heaviside functions, $H_{l,i,m}^t$ and $H_{s,i,m}^t$, are introduced in the above liquid- and solid-phase ODEs, respectively. These functions are equal to 1 (or 0) when a particular phase or species is present (or not present), and are used to force the derivatives to zero when a particular phase or species is not present or a nonzero species in a given phase is not “active.” This is discussed in more detail in subsections 4.2.1, 4.2.2, and 4.2.3.

[36] Using the approach given by Jacobson [2005], we discretize the liquid-phase ODEs (equation (15)) to obtain the following semi-implicit Euler expression:

$$\frac{C_{l,i,m}^{t+h} - C_{l,i,m}^t}{h} = k_{i,m} H_{l,i,m} \left(C_{g,i}^{t+h} - C_{l,i,m}^{*,t+h} \right) \quad (17)$$

where superscripts t and $t + h$ represent the old and new times, respectively, and h is the time step taken at time t to advance to $t + h$. All the variables at time t are known either from initial conditions (i.e., at $t = 0$) or the previous time step while the variables at time $t + h$ are unknown. Equilibrium surface concentration of a species over the liquid-phase can be expressed in terms of its liquid-phase concentration as:

$$C_{l,i,m}^{*,t+h} = S_{i,m}^t C_{l,i,m}^{t+h} \quad (18)$$

where $S_{i,m}^t$ is the saturation ratio of species i in aerosol bin m . Note that its value is evaluated at time t instead of $t + h$, which makes equation (17) semi-implicit as opposed to fully-implicit. Substituting equation (18) in (17) and solving it for the new liquid-phase concentration, $C_{l,i,m}^{t+h}$ gives:

$$C_{l,i,m}^{t+h} = \frac{C_{l,i,m}^t + h k_{i,m}^t H_{l,i,m} C_{g,i}^{t+h}}{1 + h k_{i,m}^t H_{l,i,m} S_{i,m}^t}. \quad (19)$$

[37] It is not possible to derive a similar semi-implicit discrete equation for the solid-phase ODEs, because the equilibrium surface concentration of a species over the solid-phase cannot be expressed in terms of its solid-phase concentration. We therefore discretize the solid-phase ODEs (equation (16)) using the explicit Euler method:

$$\frac{C_{s,i,m}^{t+h} - C_{s,i,m}^t}{h} = H_{s,i,m} k_{i,m}^t (C_{g,i}^t - C_{s,i,m}^{*,t}) = J_{s,i,m}^t \quad (20)$$

where $J_{s,i,m}^t$ is the flux of species i to the solid-phase in bin m . Note that all the terms on the right-hand-side (RHS) are evaluated (and explicitly known) at time t . Solving equation (20) for the new solid-phase species concentration, $C_{s,i,m}^{t+h}$, then simply gives:

$$C_{s,i,m}^{t+h} = C_{s,i,m}^t + h J_{s,i,m}^t. \quad (21)$$

Now, the overall mass balance for a given species i present in the gas-phase and in both the solid and liquid phases over all the size bins can be written as:

$$C_{g,i}^{t+h} + \sum_m C_{s,i,m}^{t+h} + \sum_m C_{l,i,m}^{t+h} = T_i. \quad (22)$$

where T_i is the total gas- plus particle-phase species concentration, which is always known at the beginning of each adaptive time step. Substituting equations (19) and (21) in (22) and solving it for the new gas-phase concentration, $C_{g,i}^{t+h}$, gives:

$$C_{g,i}^{t+h} = \frac{T_i - \sum_m \left[\left(C_{s,i,m}^t + h J_{s,i,m}^t \right) - \left(\frac{C_{l,i,m}^t}{1 + h k_{i,m}^t S_{i,m}^t H_{l,i,m}^t} \right) \right]}{1 + h \sum_m \left(\frac{k_{i,m}^t H_{l,i,m}^t}{1 + h k_{i,m}^t S_{i,m}^t H_{l,i,m}^t} \right)}. \quad (23)$$

Note that all the terms on the RHS of the above equation are evaluated at time t and, therefore, known. The problem now reduces to the determination of $J_{s,i,m}^t$, $S_{i,m}^t$, $H_{i,i,m}^t$, and h in a manner that produces smooth, accurate, and efficient solutions at all RH. Algorithms for calculating these terms at each time step are described below for completely solid, completely liquid, and mixed-phase particles. Once these terms are calculated, the new gas-phase species concentrations are computed with equation (23), and the new solid- and liquid-phase species concentrations are then computed with equations (19) and (21), respectively. At the end of each time step, the updated solid- and liquid-phase concentrations are summed up to compute the total particle-phase concentrations and are also input to the thermodynamic module to update the intraparticle phase equilibrium. This procedure is repeated until the end of the time splitting interval h_{ASTEM} is reached. Also, the entire ASTEM algorithm is repeated until the end of the transport-chemistry time splitting interval h_{CHEM} is reached. Readers who would prefer to forego the details for determining $J_{s,i,m}^t$, $S_{i,m}^t$, $H_{i,i,m}^t$, and h may skip to section 5 on results and discussion.

4.2.1. Completely Solid Particles

[38] If the thermodynamic module predicts that the particles in bin m are completely solid at the given RH, then the gas fluxes to/from solid particles are determined using one of the following four “solid cases.” The criterion for each case is checked in the given order. If it is satisfied for a particular case, then the procedure given for that case is used to compute $J_{s,i,m}^t$. If it is not satisfied, then the criterion for the next case is checked. If none of the criteria is met then $J_{s,i,m}^t$ for all the species is set to zero. Since the liquid-phase is not active in completely solid particles, all the liquid-phase species Heaviside functions $H_{l,i,m}^t$ are set to zero for this bin.

4.2.1.1. Solid Case 1

[39] *Criterion:* $n_{s,\text{CaCO}_3,m}^t > 0$

[40] Here, n is the concentration of salts in mol m^{-3} (air). Since CaCO_3 is sparingly soluble, it is treated as a solid in MESA over the entire RH range to increase computational efficiency [Zaveri *et al.*, 2005b]. When present in the aerosol, it reacts irreversibly with HNO_3 and HCl gases to form particulate $\text{Ca}(\text{NO}_3)_2$ and CaCl_2 , respectively (reactions (R12) and (R15) in Table 1). Since these reactions are assumed to proceed to completion, the equilibrium surface concentrations of HNO_3 and HCl are equal to zero until all $\text{CaCO}_3(\text{s})$ is exhausted. Moreover, NH_3 is not allowed to condense during this time. Thus the species fluxes in this case are:

$$J_{s,\text{HNO}_3,m}^t = k_{\text{HNO}_3,m}^t C_{g,\text{HNO}_3}^t, \quad (24)$$

$$J_{s,\text{HCl},m}^t = k_{\text{HCl},m}^t C_{g,\text{HCl}}^t, \quad \text{and} \quad (25)$$

$$J_{s,\text{NH}_3,m}^t = 0. \quad (26)$$

4.2.1.2. Solid Case 2

[41] *Criterion:* $X_{s,m}^t < 2$

[42] Here, $X_{s,m}^t$ is the sulfate ratio in the solid phase, which is defined as:

$$X_{s,m}^t = \frac{C_{s,\text{NH}_4,m}^t + C_{s,\text{Na},m}^t + 2C_{s,\text{Ca},m}^t}{C_{s,\text{SULF},m}^t + 0.5C_{s,\text{CH}_3\text{SO}_3,m}^t}. \quad (27)$$

When $X_{s,m}^t < 2$, the particle-phase is said to be “sulfate rich.” Under this condition, there are excess sulfate and/or methanesulfonate equivalent moles compared to the sum of all the equivalent cation moles, except H^+ . As a result, the particle is highly acidic and can absorb any available NH_3 , but none of the acid gases. The irreversible reactions applicable to this case include (R16)–(R18) in Table 1. Thus the species fluxes in the present case are:

$$J_{s,\text{HNO}_3,m}^t = 0, \quad (28)$$

$$J_{s,\text{HCl},m}^t = 0, \quad \text{and} \quad (29)$$

$$J_{s,\text{NH}_3,m}^t = k_{\text{NH}_3,m}^t C_{g,\text{NH}_3}^t. \quad (30)$$

4.2.1.3. Solid Case 3

[43] This case has two sub cases.

4.2.1.3.1. Solid Case 3a

[44] *Criterion:* $(n_{s,\text{NaCl},m}^t + n_{s,\text{CaCl}_2,m}^t) > 0$ and $C_{g,\text{HNO}_3}^t > 0$

[45] Solid NaCl and CaCl_2 are assumed to react irreversibly with gaseous HNO_3 to form solid $\text{NaNO}_3(\text{s})$ and $\text{Ca}(\text{NO}_3)_2(\text{s})$, respectively (reactions (R13) and (R14) in Table 1). Furthermore, these reactions are assumed to proceed to completion until all $\text{NaCl}(\text{s})$ and $\text{CaCl}_2(\text{s})$ are exhausted. The equilibrium surface concentration of HNO_3 is therefore assumed equal to zero. Also, since HCl is replaced by HNO_3 , the evaporation flux of HCl is equal to the condensation flux of HNO_3 . NH_3 is not allowed to condense or evaporate during this time unless solid case 3b is also satisfied. The species fluxes for this subcase only are:

$$J_{s,\text{HNO}_3,m}^t = k_{\text{HNO}_3,m}^t C_{g,\text{HNO}_3}^t, \quad (31)$$

$$J_{s,\text{HCl},m}^t = -J_{s,\text{HNO}_3,m}^t, \quad \text{and} \quad (32)$$

$$J_{s,\text{NH}_3,m}^t = 0. \quad (33)$$

4.2.1.3.2. Solid Case 3b

[46] *Criterion:* $n_{s,\text{NH}_4\text{Cl},m}^t > 0$ or $C_{g,\text{HCl}}^t C_{g,\text{NH}_3}^t > K_{\text{NH}_4\text{Cl}}^{\text{gs}}$

[47] In addition to the solid case 3a criterion, the above criterion must also be satisfied to enter this subcase. Under these conditions, the equilibrium reaction (E1) in Table 2 is active. Additional fluxes for HCl and NH_3 are computed

Table 2. List of Gas-Particle Equilibrium Reactions Included in MOSAIC

		Equilibrium Constant, ^{a,b} $K(T)$				
No.	Equilibrium Reaction	Symbol	$K(T_0)$	α	β	Units
<i>gas-solid</i>						
(E1)	$\text{NH}_4\text{Cl}(s) \leftrightarrow \text{NH}_3(g) + \text{HCl}(g)$	$K_{\text{NH}_4\text{Cl}}^{gs}$	8.43×10^{-17c}	-71.00	2.40	atm^2
(E2)	$\text{NH}_4\text{NO}_3(s) \leftrightarrow \text{NH}_3(g) + \text{HNO}_3(g)$	$K_{\text{NH}_4\text{NO}_3}^{gs}$	4.72×10^{-17c}	-74.38	6.12	atm^2
<i>gas-liquid</i>						
(E3)	$\text{NH}_3(g) \leftrightarrow \text{NH}_3(aq)$	$K_{\text{NH}_3}^{gl}$	57.64	13.79	-5.39	$\text{mol kg}^{-1} \text{atm}^{-1}$
(E4)	$\text{HNO}_3(g) \leftrightarrow \text{H}^+(aq) + \text{NO}_3^-(aq)$	$K_{\text{HNO}_3}^{gl}$	2.51×10^6	29.47	16.84	$\text{mol}^2 \text{kg}^{-2} \text{atm}^{-1}$
(E5)	$\text{HCl}(g) \leftrightarrow \text{H}^+(aq) + \text{Cl}^-(aq)$	K_{HCl}^{gl}	2.00×10^6	30.21	19.91	$\text{mol}^2 \text{kg}^{-2} \text{atm}^{-1}$
<i>liquid-liquid</i>						
(E6)	$\text{H}_2\text{O}(aq) + \text{NH}_3(aq) \leftrightarrow \text{OH}^-(aq) + \text{NH}_4^+(aq)$	$K_{\text{NH}_3}^{ll}$	1.81×10^{-5}	1.50	26.92	mol kg^{-1}
(E7)	$\text{H}_2\text{O}(aq) \leftrightarrow \text{H}^+(aq) + \text{OH}^-(aq)$	$K_{\text{H}_2\text{O}}^{ll}$	1.01×10^{-14}	-22.52	26.92	$\text{mol}^2 \text{kg}^{-2}$
(E8)	$\text{HSO}_4^-(aq) \leftrightarrow \text{H}^+(aq) + \text{SO}_4^{2-}(aq)$	$K_{\text{HSO}_4^-}^{ll}$	1.05 ^d	8.85	25.14	mol kg^{-1}

^aEquilibrium constant at any temperature (T) is calculated from $K(T) = K(T_0) \exp \left[\alpha \left(\frac{T_0}{T} - 1 \right) + \beta \left(1 + \ln \left(\frac{T_0}{T} \right) - \frac{T_0}{T} \right) \right]$, where the reference temperature, $T_0 = 298.15$ K.

^bThe values of equilibrium constants are from Kim *et al.* [1993] unless otherwise indicated.

^c $K(T_0)$ adjusted to reproduce the equilibrium gas concentration products predicted by the AIM (Web Model III) at 298.15 K.

^d $K(T_0)$ value is from Clegg *et al.* [1998].

using the coupled condensation relation and the respective equilibrium equations for HCl and NH_3 given below:

$$k_{\text{HCl},m}^t (C_{g,\text{HCl}}^t - C_{s,\text{HCl},m}^{*,t}) = k_{\text{NH}_3,m}^t (C_{g,\text{NH}_3}^t - C_{s,\text{NH}_3,m}^{*,t}), \text{ and} \quad (34)$$

$$C_{s,\text{NH}_3,m}^{*,t} C_{s,\text{HCl},m}^{*,t} = K_{\text{NH}_4\text{Cl}}^{gs}. \quad (35)$$

Solving equations (34) and (35) simultaneously yields a quadratic equation in $C_{s,\text{NH}_3,m}^{*,t}$, with the solution given as:

$$C_{s,\text{NH}_3,m}^{*,t} = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and} \quad (36)$$

$$C_{s,\text{HCl},m}^{*,t} = \frac{K_{\text{NH}_4\text{Cl}}^{gs}}{C_{s,\text{NH}_3,m}^{*,t}}. \quad (37)$$

where

$$a = k_{\text{NH}_3,m}^t, \quad (38)$$

$$b = k_{\text{HCl},m}^t C_{g,\text{HCl}}^t - k_{\text{NH}_3,m}^t C_{g,\text{NH}_3}^t, \text{ and} \quad (39)$$

$$c = -\left(k_{\text{HCl},m}^t K_{\text{NH}_4\text{Cl}}^{gs} \right). \quad (40)$$

[48] The net flux for HCl is then the sum of the evaporative flux (due to HNO_3 condensation) and the other flux due to gas-solid NH_4Cl equilibrium:

$$J_{s,\text{HCl},m}^t = -J_{s,\text{HNO}_3,m}^t + k_{\text{HCl},m}^t (C_{g,\text{HCl}}^t - C_{s,\text{HCl},m}^{*,t}). \quad (41)$$

The NH_3 flux is due to gas-solid NH_4Cl equilibrium only:

$$J_{s,\text{NH}_3,m}^t = k_{\text{NH}_3,m}^t (C_{g,\text{NH}_3}^t - C_{s,\text{NH}_3,m}^{*,t}). \quad (42)$$

4.2.1.4. Solid Case 4

[49] *Criterion 1:* $n_{s,\text{NH}_4\text{Cl},m}^t > 0$ or $C_{g,\text{HCl}}^t C_{g,\text{NH}_3}^t > K_{\text{NH}_4\text{Cl}}^{gs}$

[50] *Criterion 2:* $n_{s,\text{NH}_4\text{NO}_3,m}^t > 0$ or $C_{g,\text{HNO}_3}^t C_{g,\text{NH}_3}^t > K_{\text{NH}_4\text{NO}_3}^{gs}$

[51] In this case, equilibrium reactions (E1) and/or (E2) in Table 2 may be active. If only Criterion 1 is satisfied then only (E1) is active. The fluxes for HCl and NH_3 are then obtained using the coupled condensation technique described in “solid case 3b,” but excluding the HNO_3 condensation term in equation (41). Similarly, if only Criterion 2 is satisfied then only (E2) is active and the fluxes for HNO_3 and NH_3 are obtained in a similar fashion as well (i.e., using coupled condensation for HNO_3 and NH_3 along with the equilibrium equation for (E2)).

[52] However, if both the criteria are satisfied, then both (E1) and (E2) are active. In this case, the HNO_3 and HCl fluxes are still estimated as described above, independent of each other. The net NH_3 flux is then computed from the sum of the acid fluxes such that it does not exceed the maximum NH_3 flux (i.e., with $C_{s,\text{NH}_3,m}^{*,t} = 0$):

$$J_{s,\text{NH}_3,m}^t = \min \left((J_{s,\text{HNO}_3,m}^t + J_{s,\text{HCl},m}^t), k_{\text{NH}_3,m}^t C_{g,\text{NH}_3}^t \right). \quad (43)$$

Now, if $(J_{s,\text{HNO}_3,m}^t + J_{s,\text{HCl},m}^t) > k_{\text{NH}_3,m}^t C_{g,\text{NH}_3}^t$, then the HNO_3 and HCl fluxes are reduced by multiplying them with the factor $(k_{g,\text{NH}_3} C_{g,\text{NH}_3}^t) / (J_{s,\text{HNO}_3,m}^t + J_{s,\text{HCl},m}^t)$ so that the adjusted total acid flux is equal to the NH_3 flux computed by equation (43).

4.2.2. Completely Liquid Particles

[53] If the thermodynamic module predicts that the particles in bin m are completely liquid at the given RH, then $S_{i,m}^t$ and $H_{i,m}^t$ for HNO_3 , HCl, and NH_3 are determined using the gas-liquid and liquid-liquid equilibrium reactions

(E3)–(E8) in Table 2. Additionally, mass transfer to/from liquid particles must also take into account the Kelvin Effect term for semivolatile gases:

$$\kappa_{i,m} = \exp\left(\frac{2v_i\sigma_{l,m}}{\Re TR_{p,m}}\right) \quad (44)$$

where the partial molar volumes (v_i) for HNO_3 , HCl , and NH_3 are 31.45, 20.96, and 24.05 $\text{cm}^3 \text{mol}^{-1}$, respectively, calculated using density data given by Lide [2003], and $\sigma_{l,m}$ (dyne cm^{-1}) is the liquid-phase surface tension. The Kelvin Effect terms are computed at the beginning of the time-split interval h_{ASTEM} and assumed to remain constant over it.

[54] The general expressions for the saturation ratios $S_{i,m}^t$ for HNO_3 , HCl , and NH_3 are expressed, respectively, as:

$$S_{\text{NH}_3,m}^t = \left(\frac{\kappa_{\text{NH}_3,m}}{W_m^t m_{\text{H}^+,m}^t}\right) \left(\frac{\gamma_{\text{NH}_4\text{HSO}_4,m}^t}{\gamma_{\text{HHSO}_4,m}^t}\right) \left(\frac{K_{\text{H}_2\text{O}}^{\text{ll}}}{K_{\text{NH}_3}^{\text{ll}} K_{\text{NH}_3}^{\text{gl}}}\right) \quad (45)$$

$$S_{\text{HNO}_3,m}^t = \frac{\kappa_{\text{HNO}_3,m} m_{\text{H}^+,m}^t (\gamma_{\text{HNO}_3,m}^t)^2}{W_m^t K_{\text{HNO}_3}^{\text{gl}}}, \quad \text{and} \quad (46)$$

$$S_{\text{HCl},m}^t = \frac{\kappa_{\text{HCl},m} m_{\text{H}^+,m}^t (\gamma_{\text{HCl},m}^t)^2}{W_m^t K_{\text{HCl}}^{\text{gl}}} \quad (47)$$

[55] The activity coefficients (γ) needed in the above equations are noniteratively determined as described by Zaveri *et al.* [2005a]. Furthermore, the Heaviside function $H_{l,i,m}^t$ for a given species is set to unity if the species concentration in the gas- or particle-phase is nonzero; otherwise it is set to zero. What remains to be determined is the H^+ ion molality, $m_{\text{H}^+,m}^t$. As discussed at length in section 3, one of the main sources of instability and error in the numerical solution for gas-liquid partitioning is the independent transfer of the acids and NH_3 , which causes the equilibrium H^+ ion concentration to oscillate. This long-standing problem is solved here with the introduction of a new concept of *dynamic* m_{H^+} (or equivalently, *dynamic* pH). The procedure for estimating the dynamic m_{H^+} differs for the following three possible “liquid cases.” Again, the criterion for each case is checked in the order given. If it is satisfied for a particular case, then the procedure given for that case is used to compute m_{H^+} . If it is not satisfied, then the criterion for the next case is checked. If none of the criteria is met then $S_{i,m}^t$ and $H_{l,i,m}^t$ are set to zero. Since the solid-phase is not active in a completely liquid bin, all the solid-phase species fluxes ($J_{s,i,m}^t$) are set to zero.

4.2.2.1. Liquid Case 1

[56] *Criterion:* $X_{l,m}^t \leq 2$

[57] Here $X_{l,m}^t$ is calculated in the same way as in equation (27), but with liquid-phase species concentrations. When the above criterion is satisfied the liquid-phase is highly acidic and the H^+ ion concentrations are comparable to the concentrations of other cations present in the system. As a result, the characteristic timescale of H^+ , as defined previously by equation (7), is reasonably long. Also, because

of high acidity, the sulfate-rich liquid phase tends to absorb negligible amounts of HNO_3 and HCl and thereby dampens the oscillatory behavior of H^+ ion concentration. Therefore, in this case, it is safe to use the internal equilibrium $m_{\text{H}^+,m}^t$ in equations (45), (46), and (47) to estimate the saturation ratios for NH_3 , HNO_3 , and HCl , respectively. The procedure for accurately and noniteratively calculating the equilibrium m_{H^+} in sulfate-rich systems is given in section 2.2.2 of Zaveri *et al.* [2005a].

4.2.2.2. Liquid Case 2

[58] *Criterion:* $(C_{\text{g},\text{NH}_3}^t + C_{l,\text{NH}_4,m}^t) > 0$

[59] Unlike in sulfate-rich systems, using the internal equilibrium m_{H^+} in sulfate-poor aerosols produces oscillations in the condensation and evaporation of HNO_3 , HCl , and NH_3 , while keeping it fixed yields incorrect steady-state results. Here, we solve this problem with the new concept of dynamic m_{H^+} , which is a function of equilibrium constants, mass transfer coefficients, and the gas- and particle-phase concentrations of all the involved species.

[60] The surface equilibrium concentrations at the old time t are expressed as:

$$C_{l,\text{HNO}_3,m}^{*,t} = \kappa_{\text{HNO}_3,m} \left(\frac{m_{\text{H}^+,m}^t m_{\text{NO}_3^-,m}^t (\gamma_{\text{HNO}_3,m}^t)^2}{K_{\text{HNO}_3}^{\text{gl}}} \right), \quad (48)$$

$$C_{l,\text{HCl},m}^{*,t} = \kappa_{\text{HCl},m} \left(\frac{m_{\text{H}^+,m}^t m_{\text{Cl}^-,m}^t (\gamma_{\text{HCl},m}^t)^2}{K_{\text{HCl}}^{\text{gl}}} \right), \quad \text{and} \quad (49)$$

$$C_{l,\text{NH}_3,m}^{*,t} = \kappa_{\text{NH}_3,m} \left(\frac{m_{\text{NH}_4^+,m}^t}{m_{\text{H}^+,m}^t} \right) \left(\frac{\gamma_{\text{NH}_4\text{NO}_3,m}^t}{\gamma_{\text{HNO}_3,m}^t} \right)^2 \times \left(\frac{K_{\text{H}_2\text{O}}^{\text{ll}}}{K_{\text{NH}_3}^{\text{ll}} K_{\text{NH}_3}^{\text{gl}}} \right). \quad (50)$$

We first eliminate $m_{\text{H}^+}^t$ in these equations by multiplying equations (48) with (50) and (49) with (50) to get the following two equations, respectively:

$$C_{l,\text{NH}_3,m}^{*,t} C_{l,\text{HNO}_3,m}^{*,t} = K_{l,\text{NH}_4\text{NO}_3,m}^{*,t}, \quad (51)$$

$$C_{l,\text{NH}_3,m}^{*,t} C_{l,\text{HCl},m}^{*,t} = K_{l,\text{NH}_4\text{Cl},m}^{*,t}. \quad (52)$$

where

$$K_{l,\text{NH}_4\text{NO}_3,m}^{*,t} = (\kappa_{\text{NH}_3,m} \kappa_{\text{HNO}_3,m}) \left(\frac{K_{\text{H}_2\text{O}}^{\text{ll}}}{K_{\text{NH}_3}^{\text{ll}} K_{\text{NH}_3}^{\text{gl}} K_{\text{HNO}_3}^{\text{gl}}} \right) \times \left(m_{\text{NH}_4^+,m}^t m_{\text{NO}_3^-,m}^t (\gamma_{\text{NH}_4\text{NO}_3,m}^t)^2 \right) \quad (53)$$

and

$$K_{l,\text{NH}_4\text{Cl},m}^{*,t} = (\kappa_{\text{NH}_3,m} \kappa_{\text{HCl},m}) \left(\frac{K_{\text{H}_2\text{O}}^{\text{ll}}}{K_{\text{NH}_3}^{\text{ll}} K_{\text{NH}_3}^{\text{gl}} K_{\text{HCl}}^{\text{gl}}} \right) \times \left(m_{\text{NH}_4^+,m}^t m_{\text{Cl}^-,m}^t (\gamma_{\text{NH}_4\text{Cl},m}^t)^2 \right). \quad (54)$$

We now use the acid-base coupled condensation approximation to close the system of algebraic equations:

$$k_{\text{HNO}_3,m}^t (C_{g,\text{HNO}_3}^t - C_{l,\text{HNO}_3,m}^{*,t}) + k_{\text{HCl},m}^t (C_{g,\text{HCl}}^t - C_{l,\text{HCl},m}^{*,t}) = k_{\text{NH}_3,m}^t (C_{g,\text{NH}_3}^t - C_{l,\text{NH}_3,m}^{*,t}) \quad (55)$$

It is worth emphasizing here that the coupled condensation approximation is used here only to estimate the dynamic $m_{\text{H}^+,m}^t$, but is not actually enforced as was done by *Wexler and Seinfeld* [1991]. Nor is the flux of H^+ ions artificially constrained with arbitrary bounds as was done by *Pilinis et al.* [2000]. Equations (51), (52), and (55) represent a system of three equations and three unknowns; solving them simultaneously, we get a quadratic equation of the form:

$$a(C_{l,\text{NH}_3,m}^{*,t})^2 + bC_{l,\text{NH}_3,m}^{*,t} + c = 0 \quad (56)$$

where

$$a = k_{\text{NH}_3,m}^t, \quad (57)$$

$$b = k_{\text{HNO}_3,m}^t C_{g,\text{HNO}_3}^t + k_{\text{HCl},m}^t C_{g,\text{HCl}}^t - k_{\text{NH}_3,m}^t C_{g,\text{NH}_3}^t, \quad (58)$$

$$c = -(k_{\text{HNO}_3,m}^t K_{l,\text{NH}_4\text{NO}_3,m}^{*,t} + k_{\text{HCl},m}^t K_{l,\text{NH}_4\text{Cl},m}^{*,t}) \quad (59)$$

The positive root of equation (56) is given by

$$C_{l,\text{NH}_3,m}^{*,t} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}. \quad (60)$$

The equilibrium surface concentrations for HNO_3 and HCl are obtained by substituting the calculated $C_{l,\text{NH}_3,m}^{*,t}$ back in equations (51) and (52), respectively. The dynamic $m_{\text{H}^+,m}^t$ is then simply estimated by rearranging equations (48) or (49) and substituting the calculated values of $C_{l,\text{HNO}_3,m}^{*,t}$ or $C_{l,\text{HCl},m}^{*,t}$, respectively:

$$m_{\text{H}^+,m}^t = \left(\frac{C_{l,\text{HNO}_3,m}^{*,t} K_{\text{HNO}_3}^{gl}}{\kappa_{\text{HNO}_3,m} m_{\text{NO}_3,m}^t (\gamma_{\text{HNO}_3,m}^t)^2} \right), \quad \text{or} \quad (61)$$

$$m_{\text{H}^+,m}^t = \left(\frac{C_{l,\text{HCl},m}^{*,t} K_{\text{HCl}}^{gl}}{\kappa_{\text{HCl},m} m_{\text{Cl}^-,m}^t (\gamma_{\text{HCl},m}^t)^2} \right). \quad (62)$$

This approach, as we will demonstrate in section 5, yields stable solutions at long time steps (~ 100 s). The dynamic $m_{\text{H}^+,m}^t$ evolves smoothly and accurately as the gas- and particle-phase concentrations of the various semivolatile and nonvolatile species change with time and eventually reach equilibrium.

4.2.2.3. Liquid Case 3

[61] *Criterion:* $(C_{g,\text{NH}_3}^t + C_{l,\text{NH}_4,m}^t) = 0$

[62] In the absence of gas- and particle-phase ammonia, the acid-base coupled condensation approximation reduces to:

$$k_{\text{HNO}_3,m}^t (C_{g,\text{HNO}_3}^t - C_{l,\text{HNO}_3,m}^{*,t}) + k_{\text{HCl},m}^t (C_{g,\text{HCl}}^t - C_{l,\text{HCl},m}^{*,t}) = 0. \quad (63)$$

Taking ratio of equations (48) and (49) eliminates $m_{\text{H}^+,m}^t$, and we get:

$$\frac{C_{l,\text{HNO}_3,m}^{*,t}}{C_{l,\text{HCl},m}^{*,t}} = r_m^{*,t} \quad (64)$$

where

$$r_m^{*,t} = \frac{\kappa_{\text{HNO}_3,m} K_{\text{HCl}}^{gl} m_{\text{NO}_3,m}^t}{\kappa_{\text{HCl},m} K_{\text{HNO}_3}^{gl} m_{\text{Cl}^-,m}^t} \left(\frac{\gamma_{\text{HNO}_3,m}^t}{\gamma_{\text{HCl},m}^t} \right)^2. \quad (65)$$

By solving equations (63) and (64) simultaneously we get

$$C_{l,\text{HCl},m}^{*,t} = \frac{k_{\text{HCl},m}^t C_{g,\text{HCl}}^t + k_{\text{HNO}_3,m}^t C_{g,\text{HNO}_3}^t}{k_{\text{HCl},m}^t + k_{\text{HNO}_3,m}^t \times r_m^{*,t}}, \quad \text{and} \quad (66)$$

$$C_{l,\text{HNO}_3,m}^{*,t} = r_m^{*,t} \times C_{l,\text{HCl},m}^{*,t}. \quad (67)$$

Now, the dynamic $m_{\text{H}^+,m}^t$ is estimated using either of the following relationships:

$$m_{\text{H}^+,m}^t = \frac{K_{\text{HNO}_3}^{gl} C_{l,\text{HNO}_3,m}^{*,t}}{\kappa_{\text{HNO}_3,m} m_{\text{NO}_3,m}^t (\gamma_{\text{HNO}_3,m}^t)^2}, \quad (68)$$

$$m_{\text{H}^+,m}^t = \frac{K_{\text{HCl}}^{gl} C_{l,\text{HCl},m}^{*,t}}{\kappa_{\text{HCl},m} m_{\text{Cl}^-,m}^t (\gamma_{\text{HCl},m}^t)^2}. \quad (69)$$

4.2.3. Mixed-Phase Particles

[63] If, at the given RH, the thermodynamic module predicts that particles in a given bin m contain both the solid and liquid phases (i.e., mixed-phase), then the following six cases are checked in the given order, and the one whose criterion is satisfied is used for determining the $J_{s,i,m}^t$, $S_{i,m}^t$, and $H_{l,i,m}^t$ terms for use in equation (23).

4.2.3.1. Mixed-Phase Case 1

[64] The criteria and procedure for this mixed-phase case is exactly the same as for “Solid Case 1.”

4.2.3.2. Mixed-Phase Case 2

[65] The criteria and procedure for this mixed-phase case is exactly the same as for “Liquid Case 1.”

4.2.3.3. Mixed-Phase Case 3

[66] The criteria and procedure for this mixed-phase case is exactly the same as for “Solid Case 3.”

4.2.3.4. Mixed-Phase Case 4

[67] This case consists of the following three subcases:

4.2.3.4.1. Mixed-Phase Case 4a

[68] *Criterion:* $n_{s,\text{NH}_4\text{NO}_3,m}^t > 0$ and $n_{s,\text{NH}_4\text{Cl},m}^t > 0$

[69] If both $\text{NH}_4\text{NO}_{3(s)}$ and $\text{NH}_4\text{Cl}_{(s)}$ are present in the solid-phase, then the procedure is exactly the same as for “Solid Case 4.”

4.2.3.4.2. Mixed-Phase Case 4b

[70] *Criterion:* $n_{s,\text{NH}_4\text{NO}_3,m}^t > 0$

[71] The procedure for this case is a combination of the “Solid Case 4” where only $\text{NH}_4\text{NO}_{3(s)}$ is active and the “Liquid Case 2” where only $\text{NH}_4\text{Cl}_{(l)}$ is active.

4.2.3.4.3. Mixed-Phase Case 4c

[72] *Criterion:* $n_{s,\text{NH}_4\text{Cl},m}^t > 0$

[73] The procedure for this case is a combination of the “Solid Case 4” where only $\text{NH}_4\text{Cl}_{(s)}$ is active and the “Liquid Case 2” where only $\text{NH}_4\text{NO}_{3(l)}$ is active.

4.2.3.5. Mixed-Phase Case 5

[74] The criterion and procedure for this mixed-phase case is exactly the same as for “Liquid Case 2.”

4.2.3.6. Mixed-Phase Case 6

[75] The criterion and procedure for this mixed-phase case is exactly the same as for “Liquid Case 3.”

4.2.4. Adaptive Time Stepping Scheme

[76] The last variable that remains to be determined in equation (23) is the time step h . Experimentation with different values for h showed that one fixed value could not ensure accuracy and computational efficiency under all possible conditions. An adaptive time stepping scheme is required. The scheme developed here adapts h to the instantaneous relative departure of the gas-phase concentrations from equilibrium with the particle-phase. At the old time t , the instantaneous relative departure from equilibrium for a gas is the same as its relative driving force, Φ , which is defined below for fluxes to the solid and liquid phases, respectively, as:

$$\Phi_{s,i,m}^t = \frac{C_{g,i}^t - C_{s,i,m}^{*,t}}{\max(C_{g,i}^t, C_{s,i,m}^{*,t})}, \quad \text{and} \quad (70)$$

$$\Phi_{l,i,m}^t = \frac{C_{g,i}^t - C_{l,i,m}^{*,t}}{\max(C_{g,i}^t, C_{l,i,m}^{*,t})}. \quad (71)$$

In this definition, the relative driving force for a gas has a value of +1 or −1 when its concentration is farthest from equilibrium, and approaches zero as it nears equilibrium. The characteristic timescale for a species is the shortest when $|\Phi| = 1$, and becomes longer as Φ approaches zero. To maintain the same level of accuracy at all times, the solver must therefore take short time steps when $|\Phi|$ is closer to unity, and increase them as Φ gradually approaches zero. This is the basis for the adaptive time stepping scheme in ASTEM. The algorithm first evaluates the adaptive time steps for each gas-phase species for both the liquid and solid phases. Because the gas-liquid ODE discretization is semi-implicit, it is possible to include both the positive and negative fluxes in adapting the gas-liquid time step to Φ without violating mass conservation:

$$h_{g \leftrightarrow l,i} = \frac{\alpha_{g \leftrightarrow l}}{\sum_m (k_{i,m} |\Phi_{l,i,m}^t|)} \quad (72)$$

where the summation is limited to bins with $|\Phi_{l,i,m}^t| > 0$. On the other hand, the explicit Euler gas-solid ODE discretization allows only the positive fluxes (i.e., condensation fluxes) to be included in adapting the gas-solid time steps to Φ without violating mass conservation:

$$h_{g \rightarrow s,i} = \frac{\alpha_{g \rightarrow s}}{\sum_m (k_{i,m}^t \Phi_{s,i,m}^t)} \quad (73)$$

where the summation is limited to bins with $\Phi_{s,i,m}^t > 0$. The procedure for handling evaporating solid-phase species will be discussed shortly. In the above equations, $\alpha_{g \leftrightarrow l}$ and $\alpha_{g \rightarrow s}$ are user-specified constants whose values must be less than or equal to 1.0. The smaller their values the more accurate the temporal solution will be. The overall maximum allowable time step h , to be used in equation (23), is then the minimum of all the individual $h_{g \leftrightarrow l,i}$ and $h_{g \rightarrow s,i}$ time steps:

$$h = \min(h_{g \leftrightarrow l,i}, h_{g \rightarrow s,i}, i = \text{HNO}_3, \text{HCl}, \text{NH}_3). \quad (74)$$

Based on tests conducted as a part of this study, a set of $\alpha_{g \leftrightarrow l} = 0.5$ and $\alpha_{g \rightarrow s} = 0.1$ appear to offer a good tradeoff between accuracy and efficiency, provided the maximum h is restricted to ~ 100 s even when the estimated adaptive value is larger.

[77] We now come back to the remaining problem of evaporating solid-phase species (i.e., $J_{s,i,m}^t < 0$). As discussed earlier, evaporation of solid-phase species is one of the sources of severe stiffness in the gas-particle partitioning ODEs. This is encountered in “Solid Case 3,” where HCl will evaporate when HNO_3 condenses on solid NaCl aerosols. Also, in “Solid Case 4,” NH_3 and HCl will evaporate from the solid phase if $n_{s,\text{NH}_4\text{Cl},m}^t > 0$ and $C_{g,\text{HCl}}^t C_{g,\text{NH}_3}^t < K_{\text{NH}_4\text{Cl}}^{\text{gs}}$. Similarly, NH_3 and HNO_3 will evaporate when $n_{s,\text{NH}_4\text{NO}_3,m}^t > 0$ and $C_{g,\text{HNO}_3}^t C_{g,\text{NH}_3}^t < K_{\text{NH}_4\text{NO}_3}^{\text{gs}}$. In such cases, the maximum time step for a given species cannot exceed $-C_{v,i,m}^t / J_{s,i,m}^t$, where $C_{v,i,m}^t$ (mol m^{−3}) is the concentration of “volatile portion” of species i in the solid-phase in bin m (i.e., one cannot evaporate more than the amount of the volatile portion of a species present in the aerosol). MESA keeps track of the concentrations of volatile solid salts such as NH_4NO_3 and NH_4Cl , which are used in ASTEM to calculate $C_{v,i,m}^t$ as:

$$C_{v,\text{NH}_4,m}^t = n_{s,\text{NH}_4\text{NO}_3}^t + n_{s,\text{NH}_4\text{Cl}}^t \quad (75)$$

$$C_{v,\text{NO}_3,m}^t = n_{s,\text{NH}_4\text{NO}_3}^t, \quad \text{and} \quad (76)$$

$$C_{v,\text{Cl},m}^t = n_{s,\text{NH}_4\text{Cl}}^t. \quad (77)$$

However, if the solid case 3a is also active, then all of the particle-phase chloride can be evaporated and is therefore considered volatile. In this case,

$$C_{v,\text{Cl},m}^t = n_{s,\text{NH}_4\text{Cl}}^t + n_{s,\text{NaCl}}^t + 2n_{s,\text{CaCl}_2}^t. \quad (78)$$

[78] Now, when the evaporating solid species concentration ($C_{v,i,m}^t$) in a particular bin is much smaller compared to the concentrations of other species in the same or other bins, then the solver would be forced to take undesirably small time steps for all the bins, and hence the stiffness. To overcome this problem, we have developed a new procedure to handle the evaporation of solid-phase species in a manner that still allows the solver to take the same time step h as calculated by equation (74). In this approach, the “individual” time step for an evaporative gas-solid flux is first calculated as a fraction ($\alpha_{s,i,m}$) of the maximum possible value as:

$$h_{s \rightarrow g,i,m} = \frac{-\alpha_{s,i,m} C_{v,i,m}^t}{J_{s,i,m}^t} \quad (79)$$

where $0 < \alpha_{s,i,m} \leq 1$. Now, if $h_{s \rightarrow g,i,m} > h$ then the evaporative gas-solid flux $J_{s,i,m}^t$ is used as is in equation (23). However, if $h_{s \rightarrow g,i,m} < h$, then $J_{s,i,m}^t$ is artificially reduced by multiplying it by $(h_{s \rightarrow g,i,m}/h)$ so that its integrated value over h is same as it would have been over $h_{s \rightarrow g,i,m}$ with the original flux. The rationale for determining the value of $\alpha_{s,i,m}$ in equation (79) is as follows. If $\alpha_{s,i,m} = 1$, then the species has the potential to be completely evaporated. This is acceptable as long as the impact on the gas-phase concentration of that species is small enough so that the driving force does not change sign in the subsequent time step. However, if the amount evaporated is enough to flip the sign over in the next time step, then a smaller value for $\alpha_{s,i,m}$ must be used to prevent an oscillatory solution. A general formula for estimating $\alpha_{s,i,m}$ is given below:

$$\alpha_{s,i,m} = \min \left(1, \frac{\left| \frac{C_{g,i}^t - C_{s,i}^{*,t}}{C_{v,i,m}^t N_{s,i,\text{evap}}^t} \right|}{1} \right) \quad (80)$$

where $N_{s,i,\text{evap}}^t$ is the total number of bins from which solid-phase species i is evaporating at old time t . $C_{g,i}^t - C_{s,i}^{*,t}$ is the weighted average driving force defined as:

$$\frac{C_{g,i}^t - C_{s,i}^{*,t}}{C_{v,i,m}^t} = \frac{\sum_m C_{v,i,m}^t (C_{g,i}^t - C_{s,i,m}^{*,t})}{\sum_m C_{v,i,m}^t} \quad (81)$$

where the summation is limited to bins with $J_{s,i,m}^t < 0$. This approach estimates the moles of a given solid species to be evaporated from each evaporating bin such that the species flux does not change sign in the next time step. However, this formula cannot always guarantee this outcome, because it does not take into account the fluxes of the same species from other liquid aerosols, if present. Nevertheless, the approach described here works very well in practice, and it significantly reduces the stiffness at moderate to low RHs where some of the available approaches are either very slow and/or would produce oscillatory solutions.

5. Results and Discussion

[79] We first evaluate MOSAIC in the box-model format with a focus on validating the new ASTEM algorithm. We then evaluate MOSAIC within PNNL's 3-D chemical trans-

port model PEGASUS using the 1987 Southern California Air Quality Study (SCAQ) data set.

5.1. Box-Model Results

[80] MOSAIC/ASTEM is evaluated here using 14 idealized test cases which represent different inorganic gas-aerosol systems commonly found in the urban, rural, and marine troposphere. The time-dependent gas and aerosol concentrations are verified against those computed with a more rigorous but computationally expensive version of the model that uses LSODES (Livermore Solver for Ordinary Differential Equations with general Sparse Jacobian matrix) [Hindmarsh, 1983] instead of ASTEM for directly integrating the gas-particle partitioning equations. Unlike ASTEM, where nonvolatile and semivolatile species integrations are time-split over h_{ASTEM} , all gas-phase species in the MOSAIC/LSODES version are integrated simultaneously. The gas-solid fluxes are computed in a similar manner as in ASTEM, with additional terms for H_2SO_4 and MSA included where necessary when calculating HNO_3 , HCl , and NH_3 fluxes. However, no constraints and/or coupled condensation approximations are imposed on the gas-liquid fluxes for any species, thus giving a highly accurate “benchmark” solution.

[81] The initial conditions for the 14 test cases are listed in Table 3. Note while the RH values may differ between the various cases, they are all at the same temperature of 298.15 K. The first 12 cases consist of monodisperse (i.e., single-bin) aerosols so that the steady-state MOSAIC results can also be validated against the equilibrium model AIM (Web-based Model III; <http://mae.ucdavis.edu/wexler/aim.html>), which serves as another independent benchmark for accuracy [Wexler and Clegg, 2002]. To make MOSAIC and AIM similar in terms of species and equilibrium reactions, precipitation of only single salts was considered in the AIM Model III runs while all the double salts and salt hydrates were excluded. We also compared the steady-state MOSAIC results and the CPU time requirements for these test cases with those obtained with the computationally efficient equilibrium model ISORROPIA [Nenes *et al.*, 1998, 1999]. Finally, test cases 13 and 14 consist of polydisperse aerosols ranging from 0.005 to 12 μm distributed over 60 logarithmically-spaced size bins. MOSAIC/ASTEM results for these two cases are verified against MOSAIC/LSODES only.

5.1.1. Model Accuracy

[82] Figure 1 shows gas and aerosol concentrations for cases 1–3. These three cases are at low RH where the given aerosols are completely solid. In case 1, the initial product of the gas-phase concentrations of HNO_3 and NH_3 is lower than $K_{\text{NH}_4\text{NO}_3}^{\text{gs}}$. As a result, some of the preexisting NH_4NO_3 rapidly evaporates until the equilibrium is reached. In case 2, solid NaCl aerosol is exposed to H_2SO_4 , HNO_3 , HCl , and NH_3 . In the first 30 min HCl is completely evaporated as H_2SO_4 and HNO_3 condense on $\text{NaCl}(s)$ to form $\text{Na}_2\text{SO}_4(s)$ and $\text{NaNO}_3(s)$, respectively. At this point, solid NH_4NO_3 formation begins as NH_3 can now condense as well. However, as H_2SO_4 continues to condense further, some $\text{NaNO}_3(s)$ is replaced by $\text{Na}_2\text{SO}_4(s)$, thus freeing up HNO_3 to form some more $\text{NH}_4\text{NO}_3(s)$. The entire system reaches a steady-state within 5 h. In case 3, solid CaCO_3 aerosol reacts with H_2SO_4 , HNO_3 , and HCl to form solid CaSO_4 ,

Table 3. Initial Conditions for Test Cases^a

Table 17. Initial Conditions for Test Cases																
No.	RH, %	Initial Gas Composition				Initial Dry Particle Size, Number, Composition, and Mass										Mass, $\mu\text{g m}^{-3}$
		H ₂ SO ₄	HNO ₃ (ppbv) ^b	HCl	NH ₃	D_{p2} , μm	N , cm^{-3}	SULF	NO ₃ ⁻	Cl ⁻	CO ₃ ⁻ ($\mu\text{mol m}^{-3}$)	NH ₄ ⁺	Na ⁺	Ca ²⁺		
Completely Solid Monodisperse Particles (Low RH Cases)																
(1)	30	0.5	2.0	0.1	0.3	0.3	1462	0.1	0.3	-	-	0.5	-	-	37.2	
(2)	30	1	15	5	20	1.0	5.08	-	-	0.1	-	-	0.1	-	5.85	
(3)	5	0.5	10	5	0.5	1.0	14.6	-	-	-	0.2	-	-	0.2	20.0	
Completely Liquid Monodisperse Particles (High RH Cases)																
(4)	85	-	0.2	0.1	0.2	0.3	42.3	0.01	-	-	-	0.005	-	-	1.065	
(5)	85	0.5	5	1	5	0.3	42.3	0.01	-	-	-	0.005	-	-	1.065	
(6)	85	0.1	0.2	0.1	0.1	0.8	9.9	-	-	0.1	-	-	0.1	-	5.85	
(7)	85	1	10	0.1	3	0.8	11.6	-	0.01	0.1	-	0.01	0.1	-	6.65	
Mixed-Phase Monodisperse Particles (Moderate RH Cases)																
(8)	55	1	10	12	25	0.5	102.4	0.1	-	-	-	0.13	-	-	12.2	
(9)	52	2.5	10	2	15	1.0	10.2	-	-	0.2	-	-	0.2	-	11.7	
(10)	52	0.5	10	5	15	1.0	7.3	-	-	-	0.1	-	-	0.1	10.0	
Monodisperse Particles (Varying RH Cases)																
(11)	30–70	1	15	5	20	1.0	5.08	-	-	0.1	-	-	0.1	-	5.85	
(12)	70–30	1	10	12	25	0.5	102.4	0.1	-	-	-	0.13	-	-	12.2	
Polydisperse Particles (60 bins)																
(13)	30	1	15	1	10	see Figure 6										24.1
(14)	85	1	15	1	10	same as case 13										24.1

^aPressure = 1 atm, Temperature = 298 K.^b($\mu\text{mol m}^{-3}$) = (ppbv) $\times$ 0.04089.

Ca(NO₃)₂, and CaCl₂, respectively. However, as more H₂SO₄ and HNO₃ continue to condense, they eventually replace CaCl₂(s) with more CaSO₄(s) and Ca(NO₃)₂(s) by completely evaporating HCl. This particular system reaches the steady-state in ~ 3 h. In all three solid aerosol cases, ASTEM predictions (lines) are in excellent agreement with the LSODES results (filled circles), and the final steady-state solutions for cases 1 and 2 matched the AIM equilibrium results (filled squares) almost exactly. On the other hand, the ISORROPIA equilibrium results (filled triangles) show some deviations from MOSAIC and AIM predictions in both these cases, especially in the particulate species concentrations and the particle phase state in case 2. (Case 3 steady-state results could not be compared with AIM because it does not treat calcium.) Also, despite using long time steps of 100 s at all times, ASTEM produced smooth and accurate solutions in all three solid cases.

[83] Figure 2 shows the gas and aerosol species concentrations for cases 4–7 at high RH where the given aerosols are completely deliquesced. The H⁺ ion concentration plotted in this figure is the liquid-phase dynamic value (described in section 4.2.2) rather than an H⁺ ion concentration computed from internal liquid-phase equilibrium. Furthermore, ASTEM used time steps of 100 s at all times. In case 4, a sulfate-rich aerosol is exposed to relatively small concentrations of HNO₃, HCl, and NH₃. Being highly acidic, the aerosol absorbs appreciable amounts of NH₃ but only negligible amounts of HNO₃ and HCl, as expected. The system reaches a steady-state in ~ 3 hours. However, because of limited NH₃ availability, the acidity is only partially neutralized, and the aerosol still remains as sulfate-rich at steady-state. The initial aerosol species concentrations in case 5 are the same as in case 4, but the initial gas-phase concentrations of HNO₃, HCl, and NH₃ are much higher while the initial H₂SO₄ concentration is still rela-

tively small. As a result, the sulfate-rich aerosol is not only able to absorb enough NH₃ to completely neutralize sulfate, but also neutralize significant amounts of dissolved HNO₃ and HCl. The aerosol becomes sulfate-poor within the first 30 min and reaches a steady-state in ~ 2 h.

[84] In case 6, a completely deliquesced NaCl aerosol is exposed to relatively small concentrations of H₂SO₄, HNO₃, HCl, and NH₃ gases. Under these conditions an appreciable amount of HNO₃ gradually dissolves into the aerosol by displacing HCl while relatively much smaller amount of NH₃ enters the particle-phase. The aerosol remains sulfate-poor throughout, and the overall gas-aerosol system reaches a steady-state in ~ 4 h. In case 7, a completely liquid NaCl aerosol containing small amounts of NH₄NO₃ is exposed to high concentrations of H₂SO₄, HNO₃, and NH₃. As a result, almost 90% of the aerosol chloride is quickly evaporated in the form of HCl as H₂SO₄ and HNO₃ condense. An appreciable amount of NH₃ also dissolves into the particle-phase to offset some of the acidity. Here, the gas-aerosol system reaches a steady-state in ~ 2.5 h.

[85] Again, in the above four liquid cases, the time-varying ASTEM predictions are in excellent agreement with the LSODES results, and all the steady-state concentrations agree with the AIM equilibrium results. The equilibrium results of ISORROPIA also agree very well with MOSAIC and AIM predictions except for the H⁺ ion concentrations, which were lower than the AIM values by factors ranging from 1.4 to 10. Also, ISORROPIA predicted zero NO₃⁻ and Cl⁻ in case 4 and HNO₃ in case 6 while small nonzero values are predicted by both MOSAIC and AIM. All the temporal MOSAIC/ASTEM solutions, including the H⁺ ion concentrations, are smooth and stable despite using long time steps of 100 s at all times. These results validate the new dynamic H⁺ concept introduced here to

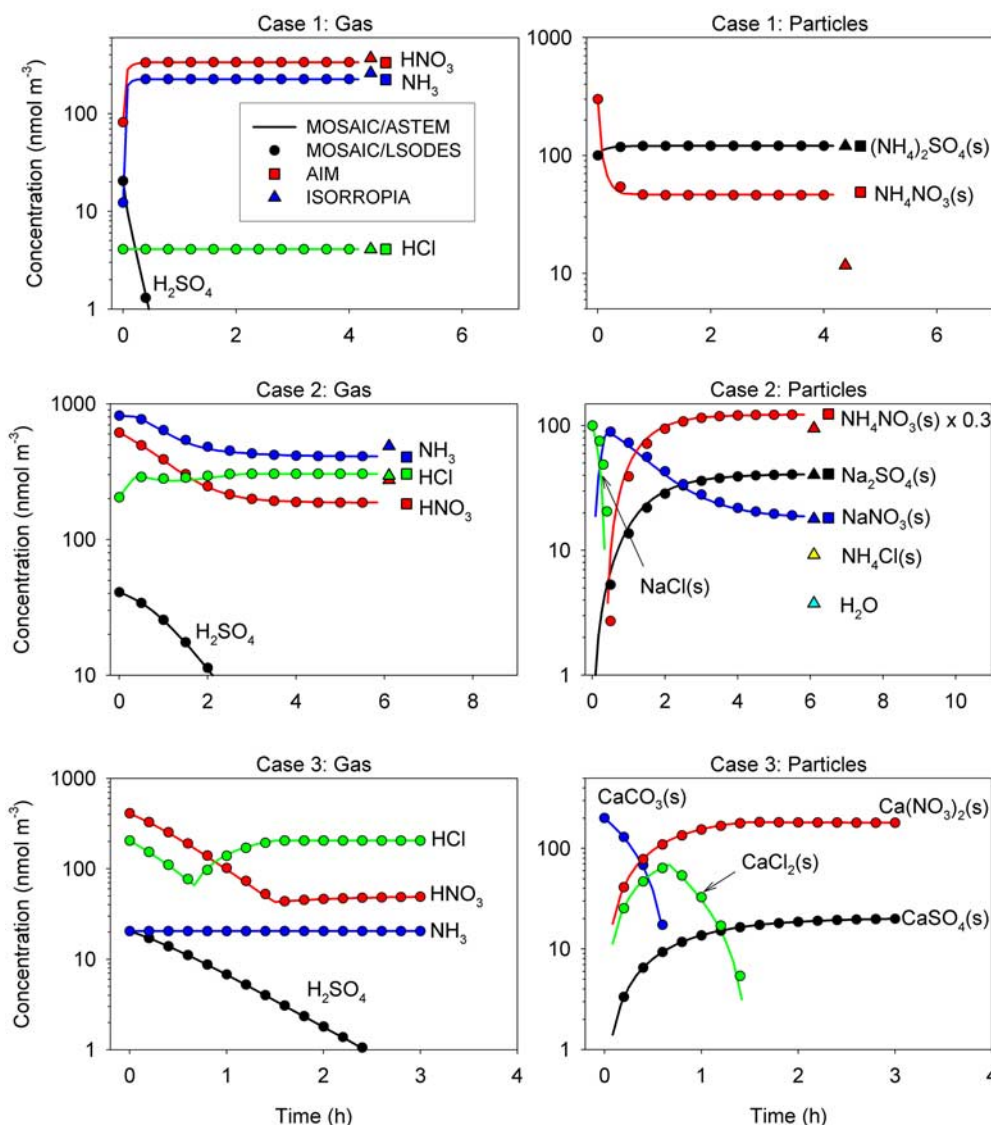


Figure 1. Comparison of the predicted gas and particles species from MOSAIC/ASTEM (lines), MOSAIC/LSODES (filled circles), equilibrium ISORROPIA (filled triangles), and equilibrium AIM Model III (filled squares) for completely solid aerosol particles test cases 1–3. Note that AIM predictions are not included for case 10 which contains calcium.

eliminate the stiffness and instability associated with the independent transfer of acids and NH_3 to aqueous aerosols.

[86] Figure 3 shows the results for cases 8–10 at moderate RH where the aerosols are composed of both the solid and liquid phases (mixed-phase). Here too, ASTEM used time steps of 100 s at all times. In case 8, a mixed-phase sulfate-rich aerosol is exposed to appreciable concentrations of H_2SO_4 , HNO_3 , HCl , and NH_3 . The aerosol quickly becomes sulfate poor as can be seen by the rapid drop in the initial H^+ ion concentration, which is followed by a gradual increase as more HNO_3 and HCl continue to condense. Under the given conditions for this case, the ions in the liquid phase are in internal equilibrium with $(\text{NH}_4)_2\text{SO}_4(\text{s})$ and $\text{NH}_4\text{Cl}(\text{s})$, and the entire system reaches a steady-state in ~ 1.5 h. In case 9, solid NaCl is exposed to appreciable concentrations of H_2SO_4 , HNO_3 , HCl , and NH_3 . Initially HCl is evaporated as H_2SO_4 and HNO_3

condense on the solid NaCl to form solid Na_2SO_4 and NaNO_3 , respectively. However, after ~ 30 min a small amount of solid NH_4Cl also forms while the solid NaCl is nearly exhausted. At this point, the composition of the particle is such that its MDRH drops below the ambient RH, and so it partially deliquesces. This dissolves some HCl back into the liquid phase. As more acids and NH_3 condense and dissolve, $\text{Na}_2\text{SO}_4(\text{s})$ concentration continues to grow. On the other hand $\text{NaNO}_3(\text{s})$ slowly dissolves and is completely lost from the solid phase after ~ 1 h. However, as more H_2SO_4 and NH_3 continue to condense, $(\text{NH}_4)_2\text{SO}_4(\text{s})$ begins to precipitate after 2.8 h and continues to slowly grow until the entire gas-aerosol system reaches a steady-state at ~ 5 h. This example illustrates that under certain conditions an apparently simple aerosol system may evolve through rather complex compositions and phase

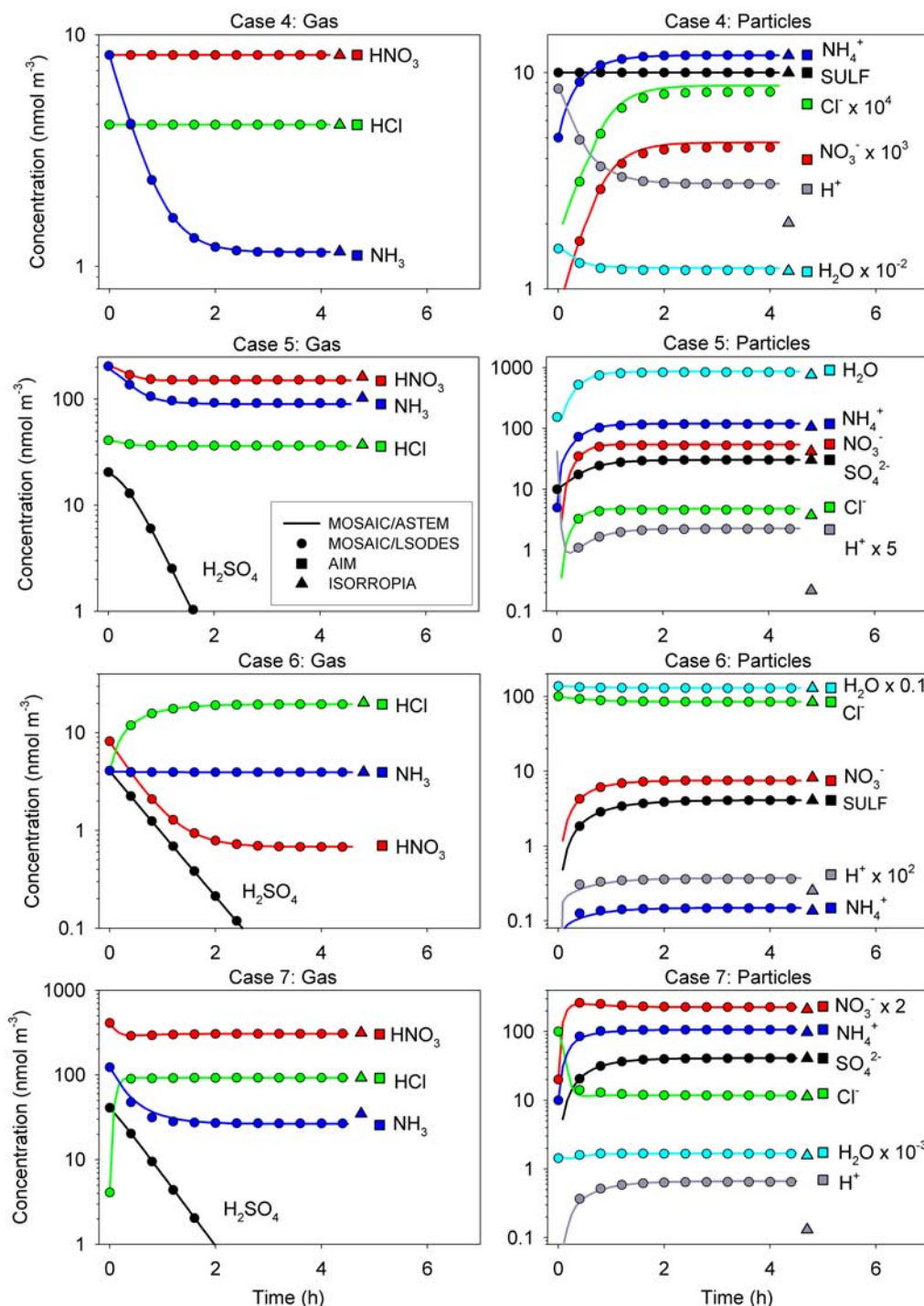


Figure 2. Comparison of the predicted gas and particles species from MOSAIC/ASTEM (lines), MOSAIC/LSODES (filled circles), equilibrium ISORROPIA (filled triangles), and equilibrium AIM Model III (filled squares) for completely liquid aerosol particles test cases 4–7.

transitions over a period of several hours before the final equilibrium is achieved.

[87] In case 10, solid CaCO_3 aerosol rapidly reacts with H_2SO_4 , HNO_3 , and HCl to form $\text{CaSO}_4(s)$, $\text{Ca}(\text{NO}_3)_2(l)$, and $\text{CaCl}_2(l)$ at the given RH. Solid CaCO_3 is completely exhausted within 30 min, at which point NH_3 begins to dissolve and precipitate along with HNO_3 and HCl to form solid NH_4NO_3 and NH_4Cl , respectively, while maintaining

the internal solid-liquid equilibrium. The entire gas-aerosol system reaches a steady-state in ~ 4 h.

[88] In the above three mixed-phase cases, the temporal ASTEM predictions are smooth and stable despite using 100 s time steps at all times. The MOSAIC/LSODES version had difficulty converging in the mixed-phase cases and therefore it could not be used to verify ASTEM. However, in cases 8 and 9, the steady-state gas, liquid,

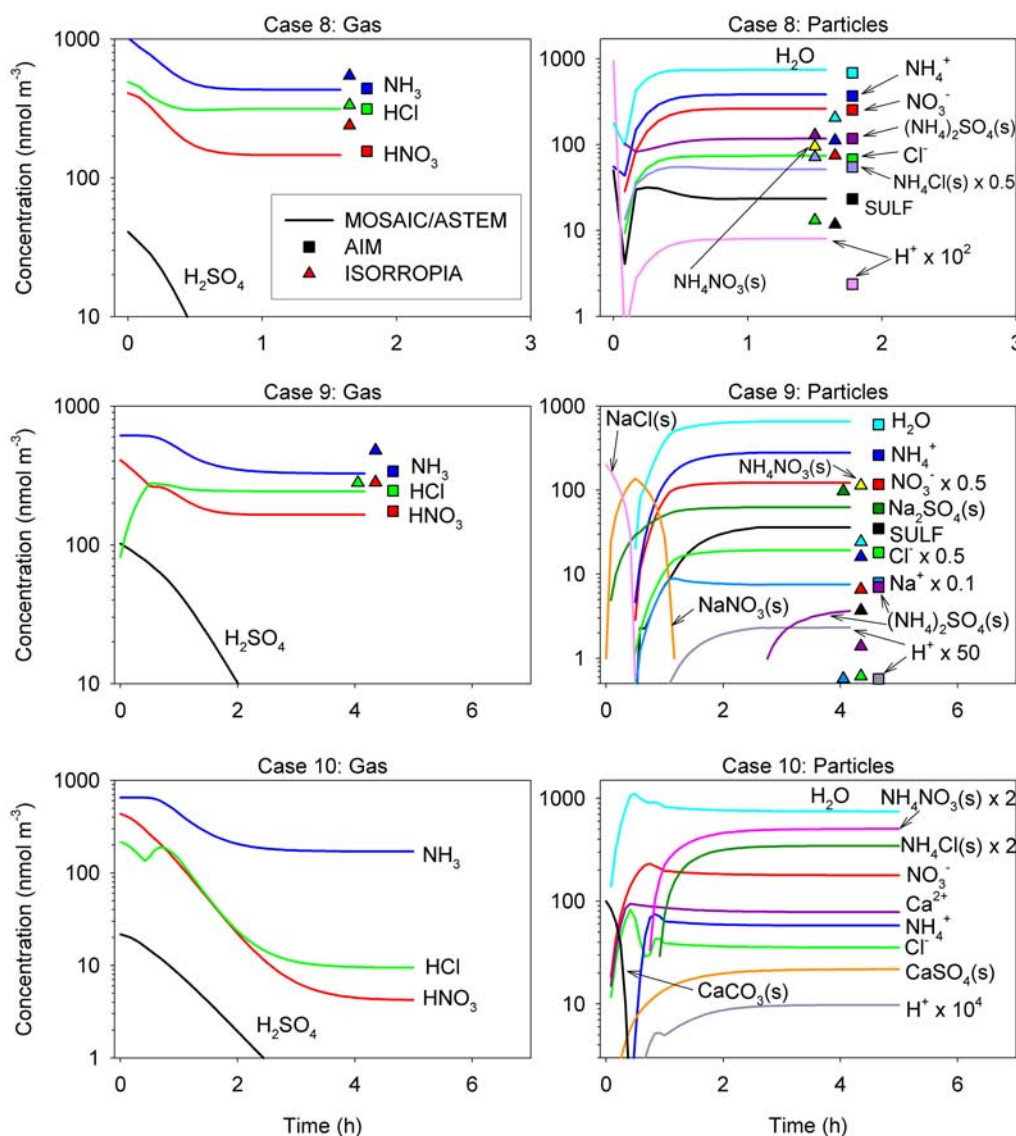


Figure 3. Comparison of the predicted gas and particles species from MOSAIC/ASTEM (lines), equilibrium ISORROPIA (filled triangles), and equilibrium AIM Model III (filled squares) for mixed-phase aerosol particles test cases 8–10. Note that ISORROPIA and AIM predictions are not included for case 10 which contains calcium.

and solid species concentrations predicted by MOSAIC/ASTEM are in excellent agreement with the AIM equilibrium results. A factor of ~ 4 discrepancy observed in the H^+ ion concentrations are due to errors in the activity coefficients predicted by MTEM with respect to the more accurate Pitzer-Simonson-Clegg (PSC) method used in AIM at moderate RH [Zaveri *et al.*, 2005a]. On the other hand, the equilibrium results from ISORROPIA show significant deviations from MOSAIC and AIM predictions in both these cases for several gases and particulate species concentrations as well as water content. For instance, it overpredicts HNO_3 and NH_3 concentrations by 55 to 60% and 25 to 42%, respectively, while underpredicts water content by 70 to 95%, with a factor of more than 3×10^5 underprediction in the H^+ ion concentrations. ISORROPIA also appears to form $\text{NH}_4\text{NO}_3(\text{s})$ salt in both the cases even though it is thermodynamically not favored under these

conditions, as confirmed by both AIM and MOSAIC predictions.

[89] Figures 4 and 5 show the results for test cases 11 and 12, respectively, in which the RH is varied between 30% and 70% during the simulation. These tests are designed to demonstrate MOSAIC's ability to accurately simulate phase transitions in aerosols due to changing meteorological conditions as a function of time. The RH is kept constant at the initial value till 6.67 h, followed by a linear change between 6.67 h and 10 h, and then again kept constant at the new value. The timing of RH change was chosen such that the gas-particle system reaches a steady state at both the initial and final RH values during the 15-h long simulation. Those results can then be compared to the AIM and ISORROPIA equilibrium predictions at both the initial and final RH values.

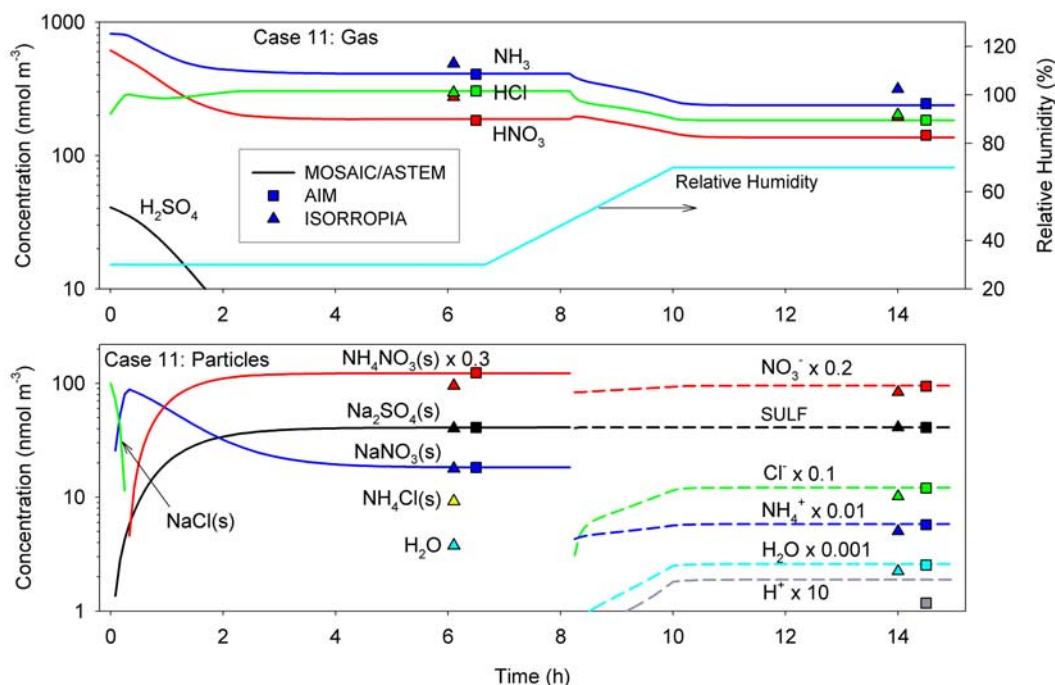


Figure 4. Comparison of the predicted gas and particles species from MOSAIC/ASTEM (lines), equilibrium ISORROPIA (filled triangles), and equilibrium AIM Model III (filled squares) for the varying RH test case 11.

[90] In test case 11, the simulation starts at 30% RH where the initial NaCl(s) aerosol particles are completely solid. The initial gas and aerosol concentrations are identical to those in test case 2. As before, the aerosol system reaches a steady-state in ~ 5 h at which time the particles are

composed of Na₂SO₄(s), NaNO₃(s), and NH₄NO₃(s), and their concentrations match the AIM equilibrium results (filled squares) almost exactly. These particles are then exposed to a linearly increasing RH from 6.67 h and 10 h such that the final RH was 70%. During this time, the

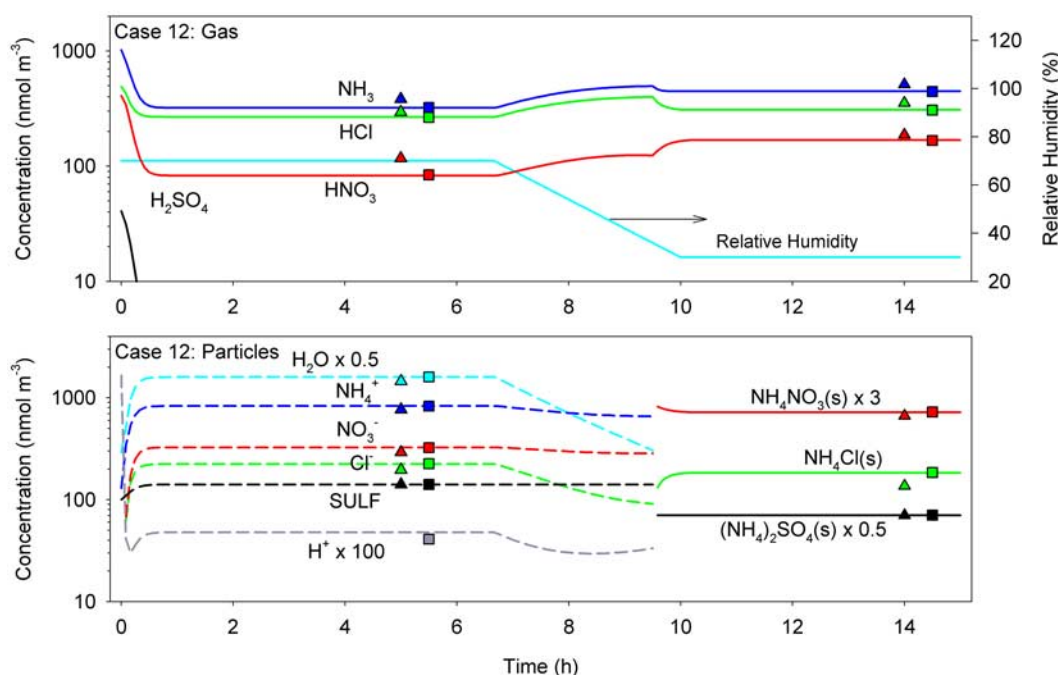


Figure 5. Comparison of the predicted gas and particles species from MOSAIC/ASTEM (lines), equilibrium ISORROPIA (filled triangles), and equilibrium AIM Model III (filled squares) for the varying RH test case 12.

Table 4. Comparison of pH Predicted by AIM (Benchmark), MOSAIC, and ISORROPIA at Equilibrium for the Monodisperse Aerosol Test Cases Containing Aqueous Phase

Test Case	AIM	MOSAIC	ISORROPIA
Case 4	−0.13	−0.14	0.03
Case 5	1.6	1.5	2.5
Case 6	3.8	3.8	4.0
Case 7	1.7	1.7	2.3
Case 8	2.7	2.2	7.7
Case 9	3.0	2.4	8.0
Case 10	−	4.1	−
Case 11	2.6	2.4	3.2
Case 12	2.2	2.1	7.9

aerosol particles were partially deliquesced when the RH reached $\sim 48\%$ and completely dissolved when RH reached 50%. Upon deliquescence and until the RH was stabilized again at 70%, the gas-particle equilibrium was continuously perturbed as the particles absorbed increasing amounts of water. Consequently, the particles continued to absorb additional amounts of NH_3 , HNO_3 , and HCl until they reached a new steady state shortly after 10 h. Again, the gas and aqueous particulate concentrations of all the species are in excellent agreement with the equilibrium values predicted by AIM. On the other hand, ISORROPIA overpredicts the equilibrium gas-phase NH_3 and HNO_3 concentrations by 30 to 50% at both the steady states.

[91] In test case 12, the initial gas and aerosol concentrations are identical to those in test case 8, except that the simulation starts at 70% RH where the initial aerosol particles are completely liquid. In this case, the liquid particles quickly absorb appreciable amounts of NH_3 , HNO_3 , and HCl , and reach a steady state within 1 h. The steady state MOSAIC results are in excellent agreement with the AIM equilibrium predictions while ISORROPIA overpredicts the gas-phase concentrations of NH_3 and HNO_3 by ~ 20 and $\sim 40\%$, respectively, and underpredicts the H^+ ion concentration by a factor of 5×10^5 . These particles are then exposed to a linearly decreasing RH from 6.67 h and 10 h such that the final RH was 30%. During this time, the aerosol particles gradually lose water until they assumed to effloresce at 35% RH. Consequently, the particles continued to gradually evaporate NH_3 , HNO_3 , and HCl , with inflexion points observed in their temporal concentration profiles at 35% RH. The completely solid particles then reach a new steady state with the gas-phase shortly thereafter. The steady state MOSAIC results are in excellent agreement with both AIM and ISORROPIA at this point.

[92] Table 4 shows the comparison of pH values predicted by AIM, MOSAIC/ASTEM, and ISORROPIA at equilibrium for the monodisperse test cases 4 through 12. MOSAIC predictions are in very good agreement with AIM while ISORROPIA predictions are generally less accurate. In the mixed-phase test cases 8, 9, and 12, ISORROPIA incorrectly predicted pH values greater than 7 while AIM and MOSAIC predicted much lower pH values between 2.2 and 3.0. The relatively larger errors in equilibrium gas- and particle-phase concentrations and the pH values predicted by ISORROPIA, particularly at low and moderate RH, primarily originate from two sources: (1) errors in the

multicomponent activity coefficients as calculated using Bromley's mixing rule at high ionic strengths [Zaveri *et al.*, 2005a], and (2) errors associated with the approximate treatment of water content and the related solid-liquid equilibria in the “mutual deliquescence region” [Nenes *et al.*, 1998]. These findings are consistent with the conclusions of Ansari and Pandis [1999] who compared another rigorous thermodynamic model, GFEMN, with ISORROPIA and found that the latter has difficulty in reproducing the complex multistage deliquescence behavior in mixed-electrolyte systems.

[93] We now evaluate MOSAIC/ASTEM for multiple size bins. Initial aerosol size and composition distributions in case 13 consist of externally-mixed $(\text{NH}_4)_2\text{SO}_4(\text{s})$ and $\text{NaCl}(\text{s})$ particles spread over 60 bins as shown in Figure 6. The $(\text{NH}_4)_2\text{SO}_4(\text{s})$ particles span the Aitken and accumulation modes (0.005 to $1 \mu\text{m}$) while the $\text{NaCl}(\text{s})$ particles are all in the coarse mode (1 to $12 \mu\text{m}$). These aerosols are exposed to high concentrations of H_2SO_4 , HNO_3 , HCl , and NH_3 at low RH (20%) such that the initial and aged particles are completely solid at all times. Figure 4 also shows the plots of predicted aerosol size and mass distributions along with species concentrations after 5 min and 10 h while the gas-phase species concentrations are shown as a continuous function of time. Mass transfer of nonvolatile species to different bins is basically governed by the relative magnitudes of their mass transfer coefficients. In the present case, the submicron aerosols, and especially the Aitken mode bins, have larger mass transfer coefficients due to greater surface area than the coarse mode bins. As a result, the submicron bins quickly grow by condensation of H_2SO_4 and NH_3 to form additional $(\text{NH}_4)_2\text{SO}_4(\text{s})$. For the same reason, greater amounts of HNO_3 and NH_3 also condense to form $\text{NH}_4\text{NO}_3(\text{s})$ on the submicron particles within the first 5 min than on the coarse $\text{NaCl}(\text{s})$ bins. H_2SO_4 and HNO_3 slowly condensing on the coarse bins form $\text{Na}_2\text{SO}_4(\text{s})$ and $\text{NaNO}_3(\text{s})$, respectively, by evaporating HCl . Because the equilibrium surface concentration of HNO_3 over solid NaCl particles is zero, it continues to condense to form $\text{NaNO}_3(\text{s})$ at the expense of $\text{NH}_4\text{NO}_3(\text{s})$ evaporated from the submicron bins. As a result the gas-phase NH_3 experiences a minimum within the first hour, after which it continues to slowly increase as more $\text{NH}_4\text{NO}_3(\text{s})$ is evaporated. After 10 h, only trace amounts of $\text{NaCl}(\text{s})$ are left in the tail end of the coarse bins while significant amounts of $\text{NH}_4\text{NO}_3(\text{s})$ are lost from the submicron bins. Here the time steps in ASTEM were 60 s at all times while the ASTEM time-split interval (h_{ASTEM}) itself was 300 s. The absolute and relative tolerances used in LSODES were rather modest: $\text{ATOL} = 1 \times 10^{-6} \text{ nmol m}^{-3}$ and $\text{RTOL} = 1 \times 10^{-3}$, respectively. While ASTEM and LSODES results are in excellent agreement across the entire size distribution after 1 and 10 h, the LSODES solution took ~ 120 times more CPU time than ASTEM.

[94] Case 14 is identical to case 13 except that the RH is increased to 85% so that the given aerosols are completely liquid. As can be seen in Figure 7, the characteristics of this case are similar to the solid case 11 but with some striking differences. For instance, here too the submicron bins grow quite rapidly within the first few minutes by absorbing significant amounts of H_2SO_4 , HNO_3 , and NH_3 , and to a relative smaller but appreciable extent HCl as well. Similar

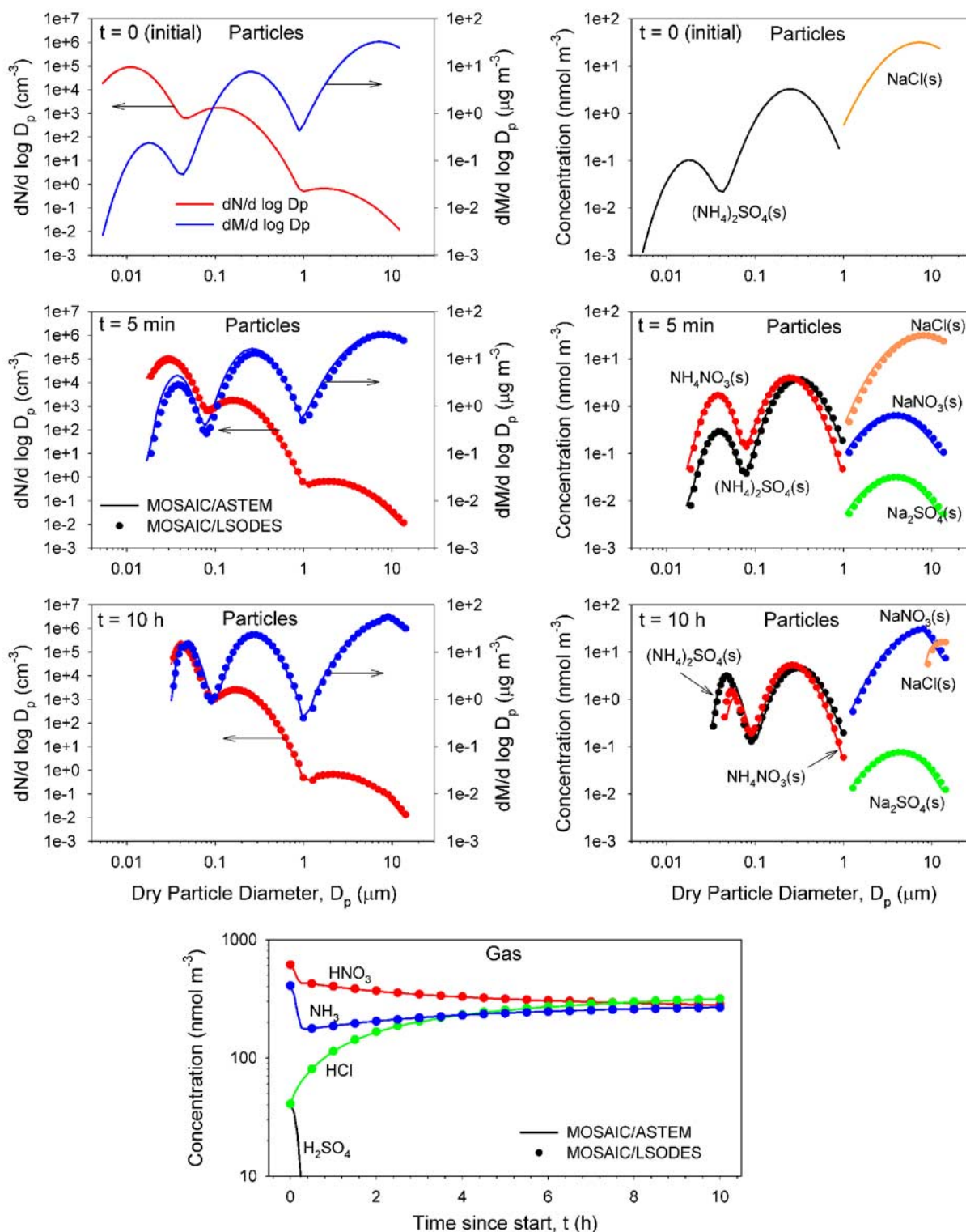


Figure 6. Comparison of the predicted gas and particles species from MOSAIC/ASTEM (lines) and MOSAIC/LSODES (filled circles) for the polydisperse aerosol test case 13 (RH = 30%, T = 298 K).

to the solid case, HNO_3 then slowly continues to dissolve into the coarse aqueous bins at the expense of the nitrate in the submicron bins. The difference is that chloride is only partially displaced by nitrate in the coarse bins, as expected in liquid sea-salt aerosols. Secondly, the NH_3 time profile

does not show a pronounced minimum, but rather a sharp drop at the beginning of the simulation, followed by a gradual decay. The slow interbin transfer of HNO_3 , HCl , and NH_3 continues to take place during this time via the gas phase but without significantly affecting their gas-phase

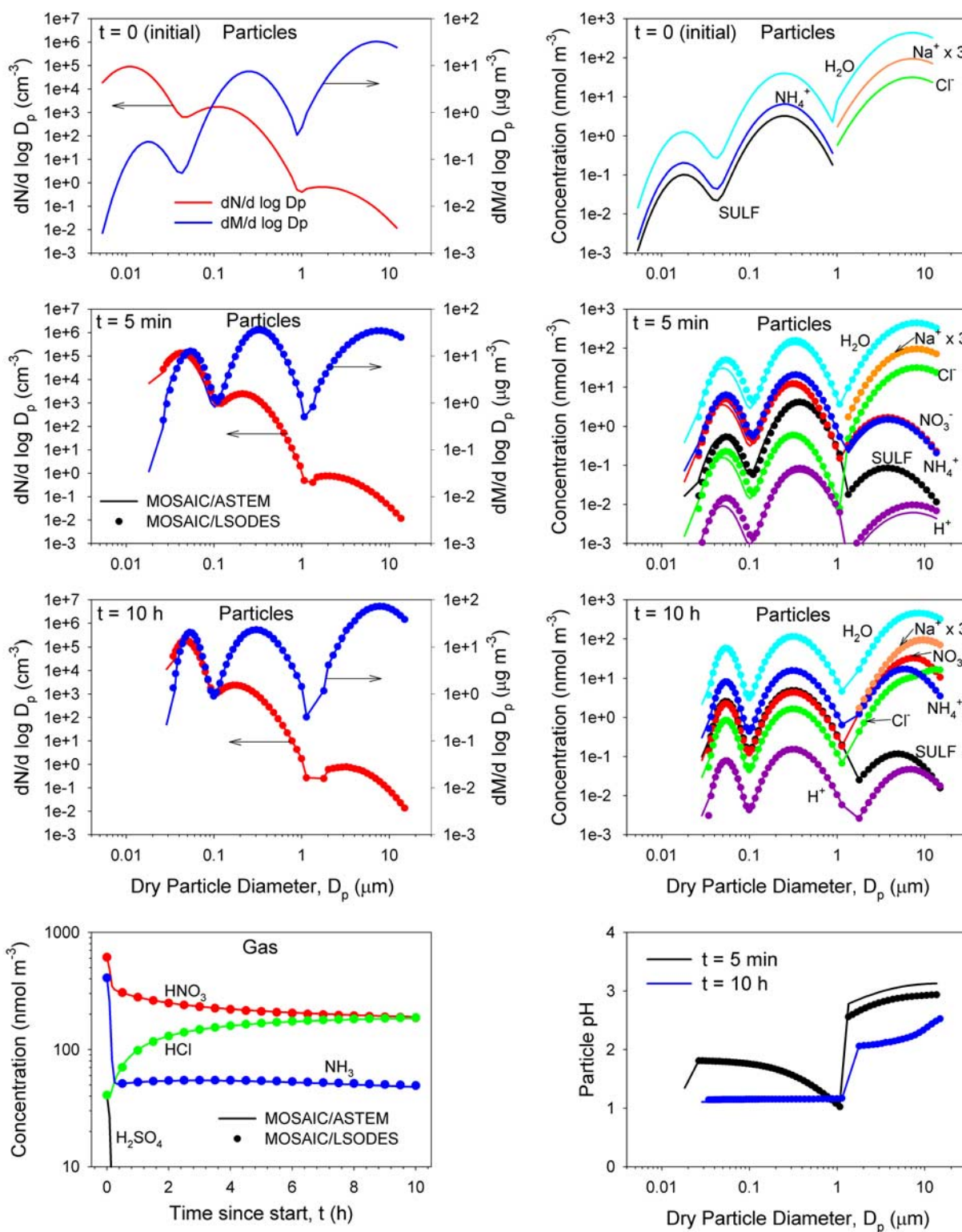


Figure 7. Comparison of the predicted gas and particles species from MOSAIC/ASTEM (lines) and MOSAIC/LSODES (filled circles) for the polydisperse aerosol test case 14 (RH = 85%, T = 298 K).

concentrations. ASTEM and LSODES results are again in excellent agreement over the entire size distribution for all the species present, including H⁺. The predicted dynamic pH shows a peculiar size and composition dependence. The

submicron aerosols which have relatively larger amounts of sulfate have significantly lower dynamic pH than the coarse Na⁺ containing aerosols. Interestingly, within the coarse aerosol bins, the dynamic pH is found to increase with size.

Table 5. Comparison of CPU Times (μ s) for MOSAIC and ISORROPIA on Different Computer Platforms and Fortran Compilers

Test Case	3.0 GHz Intel Xeon Compiler: pgf90 ^a		3.0 GHz Intel EM64T Compiler: pathf90 ^b		3.0 GHz Intel EM64T Compiler: ifort ^c		Average ^d	
	MOSAIC ^e	ISORROPIA ^f	MOSAIC ^e	ISORROPIA ^f	MOSAIC ^e	ISORROPIA ^f	MOSAIC ^e	ISORROPIA ^f
Case 1	13	3.5	13	1.9	9.3	1.8	12	2.4
Case 2	20	418	19	323	15	238	18	326
Case 3	14	-	14	-	10	-	13	-
Case 4	32	18	31	11	22	8.9	28	13
Case 5	37	52	37	45	28	34	34	44
Case 6	41	44	40	42	29	25	37	37
Case 7	40	65	41	56	29	37	37	53
Case 8	190	390	210	288	161	209	187	296
Case 9	266	107	289	86	206	57	254	83
Case 10	274	-	296	-	222	-	264	-
Case 11	15	246 ^g	15	193 ^g	11	141 ^g	14	193 ^g
Case 12	32	43 ^g	31	34 ^g	24	24 ^g	29	34 ^g
Case 13	18	593	21	474	19	355	19	474
Case 14	33	46	36	42	30	27	33	38

^aPortland Group Fortran 77/90 compiler with “-fast” optimization option.^bPathScale Fortran 77/90 compiler with “-O3” optimization option.^cIntel Fortran 77/90 compiler with “-O3” optimization option.^dAverage of CPU times obtained with the different compilers/platforms.^eMOSAIC: Average CPU time (μ s) per bin per 5 min integration interval.^fISORROPIA: CPU time (μ s) per bulk equilibrium calculation.^gAverage of the CPU times for equilibrium calculations at the initial and final RH conditions.

It is also worth noting that the dynamic pH across the entire size distribution is found to decrease with time as the gas-aerosol system approaches equilibrium.

[95] In this case too, the time steps in ASTEM were 60 s at all times. The absolute and relative tolerances used in LSODES had to be stricter at $ATOL = 1 \times 10^{-7}$ nmol m⁻³ and $RTOL = 1 \times 10^{-7}$, respectively, to obtain adequate accuracy in the presence of H⁺ ions. Again, while ASTEM and LSODES results are in excellent agreement across the entire size distribution after 5 min and 10 h, the LSODES solution in this case took ~ 320 times more CPU time than ASTEM.

5.1.2. Computational Efficiency

[96] The above analysis showed that MOSAIC/ASTEM is two orders of magnitude faster than MOSAIC/LSODES. MOSAIC's efficiency was also determined by comparing its CPU speed with that of ISORROPIA, which is considered to be one of the fastest equilibrium models currently available [Ansari and Pandis, 1999]. Since MOSAIC is a fully dynamic model, we compare its average CPU time requirement per size bin per 5 min integration interval (typical 3-D model time step) to the CPU time requirement per single, bulk equilibrium calculation with ISORROPIA. The same aerosol test cases listed in Table 3 were used for this exercise. The codes were tested with three different compilers (Portland Group Fortran 77/90, PathScale Fortran 77/90, and Intel Fortran 77/90) on two different Linux workstation platforms (3.0 GHz Intel Xeon and 3.0 GHz Intel EM64T processors). As shown in Table 5, the CPU time requirements for MOSAIC and ISORROPIA are similar for most of the cases tested in this study. However, for some test cases, MOSAIC was found to be up to 20 times faster than ISORROPIA. At the same time, MOSAIC was also found to be much more accurate than ISORROPIA under these conditions as discussed in the previous subsection. Thus, while more comprehensive comparisons of accuracy and efficiency of MOSAIC with other dynamic models are warranted in the future, the present study

demonstrates that MOSAIC is extremely efficient without sacrificing accuracy, and is therefore highly attractive for use in large scale, 3-D air quality and aerosol models.

5.2. Three-Dimensional Eulerian Model Results

[97] The above box-model tests and analysis verifies and validates the accuracy and robustness of the new thermodynamic and gas-particle partitioning modules in MOSAIC under widely different chemical and RH conditions. In this section, we evaluate MOSAIC within a comprehensive 3-D Eulerian chemical transport model, with the primary goal of estimating its computational efficiency under more realistic ambient conditions and continuously changing meteorological controls. To this end, MOSAIC was implemented in PNNL's Eulerian Gas and Aerosol Scalable Unified System (PEGASUS) [Fast et al., 2002] and applied to simulate trace gas and aerosol evolution in the Los Angeles basin between 08 UTC 26 August and 06 UTC 29 August during the 1987 Southern California Air Quality Study (SCAQs). The 1987 SCAQS data set has gained somewhat of an historic significance for evaluating new aerosol models. In addition to all the gas and aerosol processes (except nucleation and coagulation) described in section 2, emission, transport, dry deposition, and radiative transfer processes were included in this simulation. The model domain was resolved with $78 \times 28 \times 31$ grid points in the X, Y, and Z dimensions, respectively, which gives a total of 67,704 grid cells. The horizontal grid resolution was 5×5 km² while the vertical grid was nonuniformly spaced. Both the transport-chemistry time splitting interval (h_{CHEM}) and ASTEM time splitting interval (h_{ASTEM}) were set at 300 s, and the maximum allowable time step h in ASTEM was restricted to 100 s. The aerosol size distribution was resolved using 8 size bins with bin boundary diameters ranging from ~ 0.04 to 10 μ m. Three-dimensional meteorological analyses based on data interpolation and a diagnostic wind model that were used to drive the chemical transport model simulations of Meng et al. [1998] were also

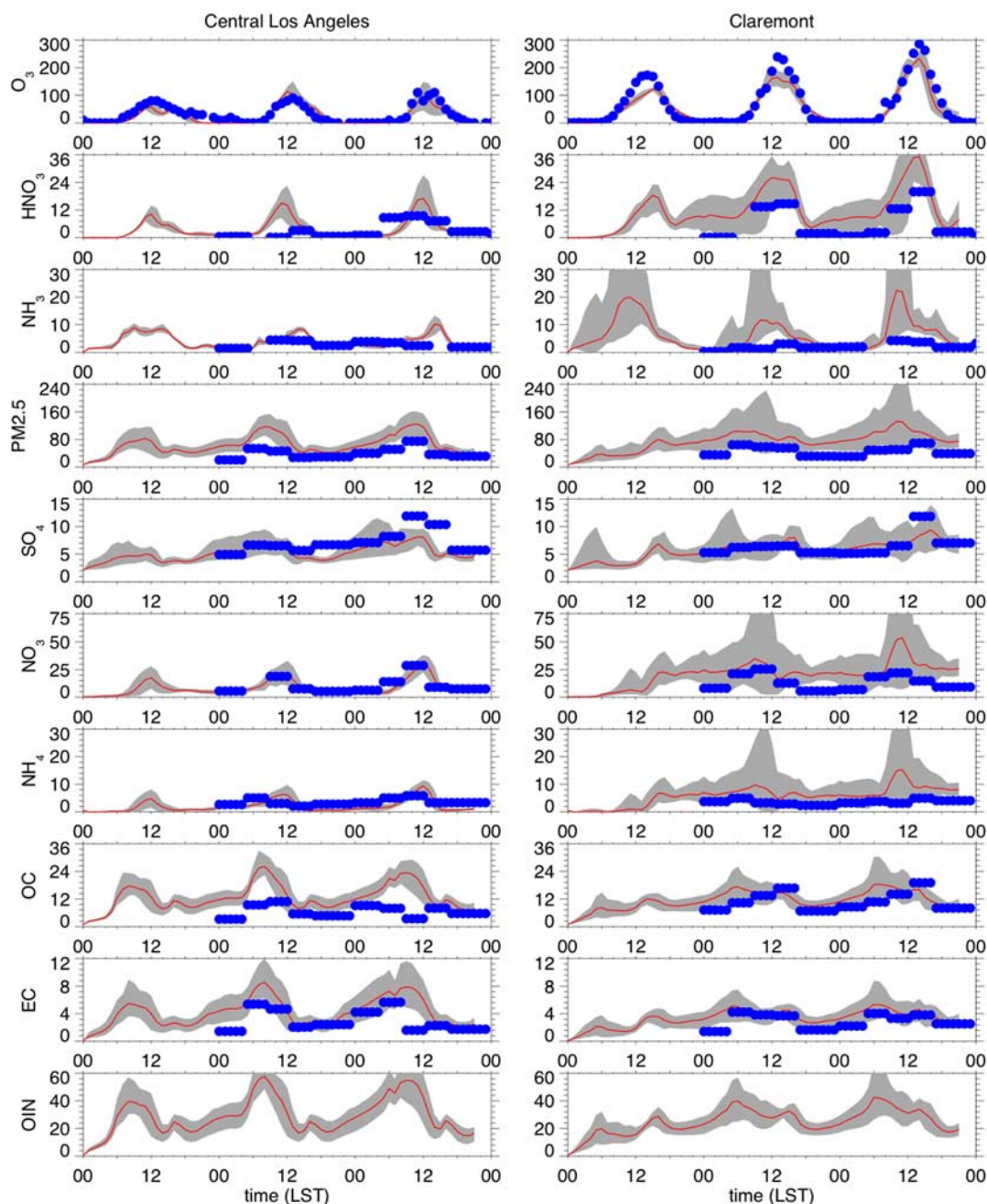


Figure 8. Comparison of the predicted (line) and observed (filled circles) diurnal variations in several key gas (ppbv) and particles ($\mu\text{g m}^{-3}$) species concentrations at the central Los Angeles and Claremont sites. The lines represent average of the values in the grid cell closest to the site and the eight horizontally-adjacent grid cells. The gray shading represents the range of the lowest and the highest values predicted in these nine grid cells.

employed in PEGASUS. Trace gas and size-resolved particulate emissions are described by Zhang *et al.* [2004] and the initial and boundary conditions were based on measurements from a remote site on San Nicolas Island, located approximately 100 km west of Los Angeles.

5.2.1. Model Evaluation

[98] Although the primary goal of this exercise is to estimate the computational efficiency of MOSAIC, it is also necessary to evaluate its performance against the available observations of the key trace gas- and particle-

phase species concentrations. While ozone was measured at many sites throughout the Los Angeles basin during the SCAQS, particulate mass and composition measurements were made at 9 sites. However, instead of carrying out an in-depth model evaluation using all of the available data, as has been done in many previous studies [Jacobson, 1997a; Meng *et al.*, 1998; Zhang *et al.*, 2004], we limit the analysis here to two representative sites for brevity. Pollutants emitted in downtown Los Angeles were transported primarily from west to east during each afternoon for 3-d period

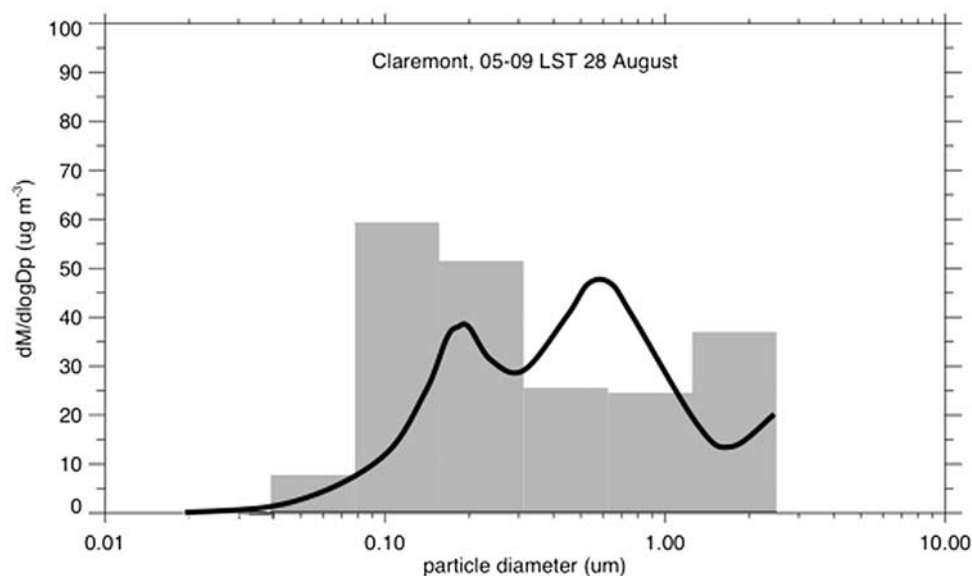


Figure 9. Comparison of the predicted (bars) and observed (line) 4-h averaged mass size distribution at Claremont. The observations are adapted from *Hering et al.* [1997].

simulated here. Because of photochemical production of secondary aerosol species, the highest particulate mass concentrations were usually measured at the downwind Claremont and Riverside sites located ~ 50 km and ~ 75 km east of downtown Los Angeles, respectively. Model performance at the central Los Angeles and Claremont sites are therefore chosen in the present evaluation.

[99] The predicted diurnal and spatial variations in ozone and particulate matter (not shown) were consistent with the observations, with the peak values east of Los Angeles in the late afternoon. Figure 8 shows comparisons of the predicted (line) and observed (filled circles) diurnal variations in several key gas and particulate (PM_{2.5}) species concentrations at the central Los Angeles and Claremont sites. The predicted values are averages of the values in the grid cell closest to the site and the eight horizontally-adjacent grid cells. The gray shading represents the range of the lowest and the highest values predicted in these nine grid cells. The predicted (average) and observed O₃ diurnal profiles are in excellent agreement, and the small differences between the minimum and maximum values in the nine grid cells indicate relatively weak horizontal gradients in O₃ concentrations at the two sites. On the other hand, relatively larger differences in the predicted minimum and maximum concentrations are seen for various aerosol species, indicating sharp gradients for aerosols at the two sites. As a result the predicted aerosol concentrations are sensitive to small errors in winds and the associated meteorological transport at these and other observation sites. Nevertheless, the predicted average concentrations for the various gas and particulate species agree quite well with the observations at the central Los Angeles and Claremont sites. The model tends to overpredict the concentrations of NH₃ and HNO₃ in the gas phase as well as particulate NH₄ and NO₃ at Claremont, although the minimum of the nine grid cells compared well with the observations. This suggests that the discrepancies between model predictions and observations at Claremont are likely due errors/uncertainties in transport

and mixing and/or in the emission inventory for NO_x and NH₃. The predicted PM_{2.5} mass was somewhat higher than observed at Los Angeles as a result of high organic carbon and other inorganic mass, while predicted PM_{2.5} was higher than observed at Claremont as a result of nitrate and other inorganic mass. Unfortunately, there were no measurements of other inorganic mass to directly evaluate the model for this species. The reason predicted organic carbon was similar to the measurements even though MOSAIC does not presently treat SOA is that emissions of organic carbon were increased from the original estimates as by *Zhang et al.* [2004].

[100] Figure 9 shows a comparison of the predicted and observed 4-h averaged aerosol mass size distribution at Claremont. A rather poor agreement between the predicted and observed mass size distribution is seen despite the use of highly accurate aerosol thermodynamics and mass transfer treatments in MOSAIC. This again suggests that the 3-D model could be missing some other process(es) that may have played an important role in shaping the observed mass size distribution at Claremont. For instance, *Koo et al.* [2003] suggest that aqueous-phase oxidation of SO₂ in cloud or fog drops may have played an important role in this episode, but these processes were not included in our simulations. Also, errors in the size distributions of the primary emitted particles can lead to even larger errors in the size distributions of the aged particles by affecting the relative rates of condensation of secondary species to the different size bins in the model.

[101] Overall, the model predictions (average and range) are in good agreement with the observations and qualitatively similar to those presented by other modeling studies [*Jacobson, 1997a; Meng et al., 1998; Zhang et al., 2004*]. Mean Normalized Gross Errors (NMGE) and Mean Normalized Biases (NMB) for 1-h averaged ozone and 24-h averaged predictions of particulate mass at all measurement sites are summarized in Table 6. The differences between PEGASUS predictions and those from other models

Table 6. Mean Normalized Gross Errors (NMGE) and Mean Normalized Biases (NMB) for 1-h Averaged Ozone and 24-h Averaged Predictions of Particulate Mass at All Measurement Sites on 27 and 28 August 1987^a

Species	SCAQS 1987 Simulation with PEGASUS (This Work)				Other SCAQS 1987 Simulations	
	27 Aug 1987		28 Aug 1987		27–28 Aug 1987	
	MNGE	MNB	MNGE	MNB	MNGE	MNB
O ₃	18.3	−15.5	22.2	−22.2	27.8–50.0	−23.0–34.0
PM _{2.5}	54.9	52.5	51.4	34.7	32.0–46.9	−8.0–46.0
PM ₁₀	95.3	92.3	87.5	75.5	50.1–72.0	−9.3–72.0
Sulfate _{2.5}	19.0	−18.5	34.6	−34.6	27.0–48.7	−30.0–48.7
Sulfate ₁₀	20.4	−20.4	36.4	−36.4	26.3–40.0	−35.4–31.6
Ammonium _{2.5}	64.1	6.3	59.2	−12.2	29.0–61.3	−52.1–61.3
Ammonium ₁₀	67.9	24.4	31.4	−19.7	23.0–49.8	−19.2–47.1
Nitrate _{2.5}	67.5	49.9	52.9	10.8	18.0–67.8	−24.8–67.8
Nitrate ₁₀	102.5	102.5	104.8	104.8	15.0–69.8	−26.6–18.4
EC _{2.5}	26.0	22.6	31.7	4.5	15.0–69.7	−23.0–35.0
EC ₁₀	30.6	19.8	23.9	−4.3	34.0–60.3	−34.1–16.2
OC _{2.5}	45.1	37.2	33.3	14.1	38.0–60.8	−44.1–14.0
OC ₁₀	144.5	136.4	102.7	91.6	32.0–64.6	−7.2–16.9
Sodium ₁₀	14.8	5.2	22.7	16.2	36.0–47.0	−35.2–38.0
Chloride ₁₀	208.0	206.3	338.4	338.4	24.0–49.2	−34.4–24.0

^aResults for other SCAQS simulations are taken from Zhang *et al.* [2004].

can be attributed to the different treatments for meteorology and aerosol formation.

5.2.2. Overall CPU Time Requirements

[102] PEGASUS was compiled with the Portland Group Fortran 77/90 compiler with “-fast” optimization and run on a single 3.0 GHz Intel Xeon processor. The entire 69 h simulation took ~25 CPU h, and the average CPU time per grid cell per hour was ~20 ms. The absolute and relative CPU time requirements for various processes are summarized in Table 7, which shows that the dynamic gas-particle partitioning, thermodynamic equilibrium, and sectional growth calculations in MOSAIC for 8 size bins took 12 ms per grid cell per hour, or nearly 60% of the total CPU time. This corresponds to an average CPU time of 125 μ s per bin per 5 min, which is comparable to the range of CPU times presented in Table 5 for the box-model tests cases.

6. Summary and Future Work

[103] We have described the various chemical, thermodynamic, and microphysical processes currently represented in the new aerosol model MOSAIC (Model for Simulating Aerosol Interactions and Chemistry), with a special focus on addressing the long-standing problems in dynamic gas-particle partitioning of semivolatile inorganic species. The coupled ordinary differential equations describing the dynamic gas-particle mass transfer process are stiff and have

proved to be extremely difficult to solve – the numerical techniques available in the literature are either computationally too expensive or produce oscillatory and/or inaccurate steady-state results. These problems were overcome in MOSAIC with a new dynamic gas-particle partitioning solver ASTEM (Adaptive Step Time-split Euler Method). The ASTEM algorithm reduces the stiffness in the ODEs by first analytically solving the condensation of all the non-volatile gases (H₂SO₄ and CH₃SO₃H) over a time-splitting interval, which is typically set at ~300 s. This is then followed by a numerical solution for the dynamic mass transfer of semivolatile gases (HNO₃, HCl, and NH₃) over the same time interval with a combination of semi-implicit and explicit Euler methods for the gas-liquid and gas-solid systems, respectively.

[104] A new concept of “dynamic pH” and an adaptive time stepping scheme were developed to further reduce the stiffness in liquid particles, and thereby allow the solver to take long time steps (~100 s) and still produce smooth and accurate solutions over the entire relative humidity range. Dynamic pH is a function of equilibrium constants, mass transfer coefficients, and the gas- and particle-phase concentrations of all the involved species, and is equal to equilibrium pH at steady state. This approach provides for the first time a more accurate calculation of aerosol pH in gas-aerosol systems at nonequilibrium compared to other

Table 7. Absolute and Relative CPU Time Requirements for MOSAIC and Other Processes Within the Off-Line 3-D Chemical Transport Model PEGASUS on a 3.0 GHz Intel Xeon Processor

Module/Model	Process	Avg. CPU Time (ms) per Grid Cell per Hour	Relative CPU Time, %
Augmented CBM-Z	Gas photochemistry	2.5	12.5
MOSAIC (8 bins)	Dynamic aerosol chemistry and microphysics	12	60
Mie & Fast-J	Aerosol optics, radiative transfer, and photolysis rates	2.5	12.5
Modified Bott's Scheme	Horizontal and vertical transport	1.8	9
Miscellaneous	Input/output and other miscellaneous calculations	1.2	6
PEGASUS	Total	20	100

formulations that are based on intraparticle thermodynamic equilibrium only. This is a significant result, because besides being important in the gas-particle partitioning of semivolatile inorganic species, particle pH also appears to play an important role in secondary organic aerosol formation via acid-catalyzed heterogeneous reactions [Jang and Kamens, 2001; Czoschke et al., 2003; Iinuma et al., 2004; Liggio et al., 2007]. Furthermore, a new method was developed that allows the solver to take long time steps even during evaporation solid-phase species that are present in low concentrations, which are a source of severe stiffness at low and moderate RH. Finally, the ASTEM algorithm simultaneously solves the appropriate gas-solid and gas-liquid mass transfer ODEs for mixed-phase particles, and thus produces accurate and smooth solutions at long time steps.

[105] MOSAIC was first evaluated for several monodisperse (single-bin) and polydisperse (60-bin) aerosol test cases representing different gas-aerosol systems commonly found in the troposphere. Performance of the new ASTEM solver was verified against a benchmark version of MOSAIC that uses the stiff ODE solver LSODES to rigorously integrate the dynamic mass transfer of all the species simultaneously without the numerical approximations used in ASTEM. The time-varying predictions with MOSAIC/ASTEM and MOSAIC/LSODES were found to be in excellent agreement for all the cases (monodisperse and polydisperse aerosols) tested under low and high RH conditions, thus validating the ASTEM algorithm. Furthermore, the final (steady-state) MOSAIC results were in excellent agreement with those obtained with the benchmark equilibrium model AIM for the monodisperse aerosol test cases at low, moderate, and high RH. On the other hand, aerosol composition/concentration, water content, and pH predicted by the computationally efficient equilibrium model ISORROPIA were also in very good agreement with AIM and MOSAIC for most of the high RH test cases, but relatively larger errors were seen under low and moderate RH conditions. Furthermore, the CPU times required for dynamic solutions by MOSAIC per size bin per 5 min integration interval (typical 3-D model time step) were similar to the CPU times required for single, bulk equilibrium solutions by ISORROPIA, although in some instances MOSAIC took ~20 times less CPU time than ISORROPIA. For the polydisperse aerosol test cases, MOSAIC/ASTEM was found to be 100–300 times faster than MOSAIC/LSODES.

[106] An 8-bin version of MOSAIC was then applied and evaluated within Pacific Northwest National Laboratory's offline 3-D chemical transport model PEGASUS for southern California area using the historic 1987 SCAQS data set. The primary goal of this exercise was to estimate MOSAIC's computational efficiency under more realistic conditions. Simulated diurnal gas and aerosol variations at the central Los Angeles and downwind Claremont sites compared well with the observations. The total CPU time per grid cell per hour averaged over the entire simulation was 20 ms (on a 3.0 GHz Intel Xeon processor), ~60% of which was used up by aerosol calculations in MOSAIC. These results show that MOSAIC is extremely efficient without compromising accuracy, and is therefore highly attractive for use in 3-D air quality and large scale aerosol models.

[107] Formation of secondary organic aerosol (SOA) and its interactions with preexisting inorganic aerosol species, primary organics, particle pH, and water are still rather poorly understood, and their impact on climate and air quality remains an area of large uncertainty in atmospheric prediction models. These processes will be included in MOSAIC in the near future as more definitive SOA mechanisms are developed and evaluated using laboratory and field data.

[108] MOSAIC is available in a standalone box-model format upon request, and is also accessible within the chemistry version of the Weather Research and Forecasting model (WRF-chem), which is publicly available.

[109] **Acknowledgments.** The authors thank A. Nenes of Georgia Tech, Atlanta, for providing a copy of the ISORROPIA code and assisting in comparing its performance with MOSAIC. The authors are grateful to D. Dabdub of University of California, Irvine, for assistance with the 1987 SCAQS data set, and A. S. Wexler of University of California, Davis, for many helpful comments on the draft manuscript. R.A.Z. is also thankful to C. M. Berkowitz and S. J. Ghan at PNNL for their support throughout this work. Funding for this research was provided by the U.S. Department of Energy (DOE) under the auspices of the Atmospheric Science Program of the Office of Biological and Environmental Research, the NASA Earth Science Enterprise under grant NAGW 3367, and Pacific Northwest National Laboratory (PNNL) Laboratory Directed Research and Development (LDRD) program. Part of the work leading to this paper was performed while R.A.Z. and L.K.P. were at Virginia Polytechnic Institute and State University, Blacksburg, Virginia. Pacific Northwest National Laboratory is operated for the U.S. Department of Energy by Battelle Memorial Institute under contract DE-AC06-76RLO 1830.

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